

Optimal Carbon Tax with Dynamic Skill Heterogeneity

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Abstract

How should a government price carbon when it also cares about redistribution? I extend the dynamic Mirrlees model to two goods, one polluting, over a life cycle of privately observed skill shocks. Under weak separability, the dirty good should be taxed at the social cost of carbon (SCC) while labor and savings distortions keep their standard forms. Only non-separability makes it nonlinear, raising it above the SCC when the dirty good is a weaker labor complement. In a US calibration with energy standing in for the dirty good however, a linear carbon tax at the SCC captures 99% of the welfare gain of an optimal nonlinear carbon tax. When labor and capital distortions are fixed at the levels implied by the US tax system, this sufficiency result is overturned as the best linear tax only captures 1.5% of the welfare gain of the best nonlinear carbon tax. Here, a linear tax at an SCC of \$190/tCO₂ would be welfare-reducing and the best linear tax would delineate to an equivalent of \$47/tCO₂. The numerical results suggest carbon tax revenue should then be recycled as a near-uniform lifetime dividend, equivalent to real-world fee-and-dividend designs.

Keywords: *Carbon tax, dynamic Mirrlees taxation, second-best environmental policy*

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1 Introduction

Carbon pricing is the economist’s favorite climate policy, and the voter’s least. France abandoned its carbon tax plans after the *gilets jaunes* protests and Canada repealed its consumer carbon tax in 2025 after it became an electoral wedge issue amid claims that it unfairly burdened households. This disconnect might in part be driven by the fact that most carbon policy is informed by integrated assessment models (Environment and Climate Change Canada (2016), Interagency Working Group on Social Cost of Greenhouse Gases (2021)), which are single-agent models that are not designed to address this concern.

Whether carbon taxes are regressive is a topic of intense study, with one aspect of the debate relating to the choice of measure. When households are ranked by annual income, taxes on gasoline and other energy appear to fall more heavily on the poor. However, when households are ranked by lifetime income, the burden appears much flatter because households smooth consumption and today’s low earners include the young and temporarily poor (Poterba 1989; Hassett et al. 2009). The welfare-relevant burden of a carbon tax is therefore a lifetime object rather than an annual one and it’s important to consider the timing of an optimal carbon tax.

However, most research focuses on the incidence of existing carbon taxes rather than asking how they should be designed in the first place. Barrage (2020) and Douenne et al. (2026) are the exceptions, but the first ties the tax system to an *ad hoc* revenue target rather than optimal redistribution and the second restricts attention to linear instruments even though income taxes are in practice nonlinear. How then should a government price carbon when it cares about redistribution, and does pollution change how income should be taxed?

The answer depends on how the income tax is modeled. The Ramsey tradition treats it as a schedule of linear instruments and usually finds that carbon should be priced below the Pigouvian level because the carbon tax compounds the distortions the rest of the tax system already imposes. The Mirrleesian tradition allows the government to levy nonlinear income taxes and usually finds that the carbon tax should be set at the Pigouvian level, unless labor supply is affected by consumption of the dirty good or by pollution.

I take the Mirrlees approach into a dynamic setting with two goods, one clean and one dirty, where agents draw privately observed skill shocks over their life cycle and their consumption of the dirty good feeds a stock externality. To my knowledge, this is the first dynamic-Mirrleesian derivation of the optimal consumer carbon tax. Compared to the static irrelevance theorems of Kaplow (2012) and Jacobs and Mooij (2015), dynamics introduce savings and history-dependence.

I find that the static irrelevance result holds in a dynamic environment. Under weak separability, the carbon wedge equals the social cost of carbon, the labor wedge keeps the Golosov et al. (2016) ABCD form, and the savings wedge keeps its inverse-Euler form. Redistributing through the optimal nonlinear income tax restores the split between corrective and redistributive taxation.

That is, the carbon price should not be adjusted to perform redistribution if nonlinear income taxes are available and optimal.

I then quantify this separation by solving the optimal mechanism, treating energy as a stand-in for the dirty good and calibrating the model to US data. I find that the optimal wedge on energy starts at the social cost of carbon and rises with the labor wedge over the life cycle, carrying a history-dependent Corlett-Hague premium that reaches about 9% by the end of working life. That premium is however small in welfare terms. When the income tax is set optimally alongside it, a linear carbon tax at the social cost of carbon already captures 99% of the welfare gain of the full optimum.

This result however assumes that every margin is set optimally. I therefore also consider the case where the carbon wedge is set when labor and savings wedges are fixed to the levels implied by the US tax system. I find that the carbon wedge is pulled down when labor is overdistorted and up when it is underdistorted. The constrained savings margin instead changes the age profile of the carbon wedge: its contribution is positive at some ages and negative at others whatever the schedule, so it cannot move the price the same way at every age. Whether the price crosses the social cost of carbon depends on the size of the swing as well as the level of the labor wedge.

Solving this constrained optimum upends the previous analysis. A linear tax at the EPA's \$190/tCO₂ social cost of carbon now shrinks consumption by 0.115% relative to having no carbon price at all. In a jointly optimized tax system, even an unoptimized linear tax set at the social cost of carbon captures 99% of the welfare gain of a nonlinear tax. In this constrained setting, the best linear tax captures only 1.5% of it. Moreover, this tax is equivalent to \$47/tCO₂, about a quarter of the social cost of carbon. These results suggest that mitigation is constrained by a suboptimal income tax system rather than by suboptimal carbon pricing design.

The solution also answers the distributional worry this paper opens with, as it suggests that carbon tax revenue should be recycled as a near-uniform lifetime dividend, equivalent to a fee-and-dividend design. The planner returns about 97% of carbon tax revenue as a lifetime transfer that is nearly identical across histories. Against carbon payments that scale with lifetime spending, the net burden falls monotonically in lifetime earnings and the bottom two lifetime-earnings quintiles gain even from a welfare-reducing linear tax at the social cost of carbon.

Section 2 reviews the literature. Section 3 presents the modeling framework. Section 4 builds the two-good dynamic Mirrlees economy and derives the optimal distortions. Section 5 studies the constrained problems: carbon pricing when the labor and savings wedges are fixed, the wedges a given carbon price induces, and the optimal linear instruments that bridge the two. Section 6 solves both cases numerically with a model calibrated to US data. Section 7 concludes.

2 Literature review

This paper sits between two literatures that reach opposite verdicts on how distortionary taxation should bend the carbon price and draws on a third that has the tools to adjudicate between them, yet has never been aimed at the environment. Should a government facing a distortionary tax system price carbon below marginal damages because the environmental tax compounds distortions elsewhere, above marginal damages because the environmental tax can be used to ease distortions elsewhere, or exactly at marginal damages?

The Ramsey tradition typically finds that the optimal carbon tax should lie below marginal damages, but that the structure of the optimal income tax is unchanged (Sandmo 1975; Bovenberg and Mooij 1994; Bovenberg and Smulders 1996). Barrage (2020) extends their static results with a dynamic climate-economy Ramsey model and finds that the optimal carbon tax should be 8-24% below the first-best Pigouvian level in the presence of distortionary income taxation. However, this analysis assumes a representative agent, focusing on efficiency and ignoring any distributional concerns.

Adding heterogeneous agents to this model overturns this result. Douenne et al. (2026) extend Barrage’s dynamic Ramsey model to heterogeneous agents, taking inspiration from the model developed in Werning (2007), and find that the optimal carbon tax is nearly Pigouvian. Douenne et al. (2025) show that this is robust to the inclusion of uninsurable income risk. Whether such a linear carbon tax is regressive and whether revenue recycling is progressive is still an open debate (Klenert et al. 2018; Goulder et al. 2019) and a parallel policy and ethics literature presses for differentiated carbon taxation (Chancel 2022; Dietsch 2024).

Without personalized carbon taxes, the exact restrictions on a uniform carbon tax reshape both its level and incidence. Using a heterogeneous agent microsimulation model calibrated to German data and focusing on revenue-neutral carbon tax reforms, Ploeg et al. (2025) find that the carbon tax should exceed marginal damages when agents have high inequality aversion. In a model calibrated to German data where households differ in income and energy efficiency, Hänsel et al. (2022) show that the first-best keeps the carbon tax uniform at the Pigouvian level and carries all redistribution through lump-sum transfers conditioned on the household’s income–energy type while leaving the income tax undistorted.

Separation can also fail for a reason orthogonal to instrument constraints. When climate damages fall unequally across households and compensation is incomplete, the social cost of carbon itself becomes distribution-dependent (Dennig et al. 2015; Kornek et al. 2021). I hold damages homogeneous to isolate the redistributive-instrument channel, so the separation I study is the Atkinson–Stiglitz one.

A static Mirrleesian literature establishes that separation, and overturns the Ramsey verdict in doing so: the below-damages result is not intrinsic to distortionary taxation but to holding the

tax system away from the optimum. Once the income tax is the optimal nonlinear schedule of Mirrlees (1971), the logic of Atkinson and Stiglitz (1976) applies and differential commodity taxes are redundant under weak separability. Cremer et al. (1998), Cremer et al. (2001), and Cremer et al. (2010) develop the environmental version of the problem where a nonlinear income tax alongside a strictly Pigouvian levy is efficient, so the public-funds correction of the Ramsey tradition disappears. Kaplow (2012) shows that any environmental reform can be made distribution-neutral by a type-by-type adjustment of the income tax, so a move to Pigouvian pricing is Pareto-improving whatever the schedule. Jacobs and Mooij (2015) show that, when goods and environmental quality are weakly separable from labor, the optimal pollution tax equals marginal damages exactly and the marginal cost of public funds is one.

When separability fails, the tax follows the logic established in Corlett and Hague (1953), which Mirrlees (1976) formalized in his model. A good is taxed above the Pigouvian level for a leisure complement, the case West and Williams (2007) substantiate for gasoline, and below it for a leisure substitute. Saez (2002) extends the benchmark to preferences that vary with ability. In a linear-tax setting, Jacobs and Ploeg (2019) show that the pollution tax stays Pigouvian when Engel curves are linear and slips below only when the poor spend disproportionately on the dirty good under nonlinear Engel curves. Every result in this literature is static, with no savings and no life-cycle dynamics. Therefore, it cannot speak to the age profile or the time path of a carbon tax.

The tools for this dynamic treatment come from the New Dynamic Public Finance literature. Golosov et al. (2003) derive a rationale for positive capital taxation in a dynamic Mirrlees economy, since it is harder to elicit future work from households that save privately. This overturns the classic zero-capital-tax benchmark common in the Ramsey tradition (Judd 1985; Chamley 1986). Moreover, the authors find that the Atkinson-Stiglitz uniform-commodity result carries over into the dynamic setting. Kocherlakota (2005) shows how to implement the capital wedge into a wealth-tax and Albanesi and Sleet (2006) derive a tractable recursive form of the problem. Kapička (2013) extends the first-order approach to persistent shocks, Pavan et al. (2014) give the general envelope characterization, Farhi and Werning (2013) characterize the labor wedge over the life cycle, and Golosov et al. (2016) extend the ABC form of the labor wedge derived in the static frameworks of Mirrlees (1971) and Saez (2001) into a dynamic ABCD form. The optimal wedges the dynamic Mirrlees model delivers vary with age, connecting to the age-dependent-tax literature (Weinzierl 2011; Bastani et al. 2013; Erosa and Gervais 2002), which prices age-varying labor rather than a commodity. None of this machinery has been aimed at commodity taxation under non-separable preferences or corrective taxation.

I extend the dynamic Mirrleesian apparatus from one consumption good to two, letting the second pollute, and characterize the optimal carbon tax alongside the labor and savings wedges over the life cycle. The heterogeneous-agent Ramsey results and this paper share the Jacobs and Mooij (2015) and Kaplow (2012) irrelevance result, which both approaches carry into a

climate economy by different routes. Douenne et al. (2026) carry it with an optimally-set linear transfer, showing analytically that the marginal cost of public funds averages to one across a balanced-growth path and equals one period by period when preferences are logarithmic. Douenne et al. (2025) find that this result still exists when agents face uninsurable labor risk. I consider uninsurable labor risk with a fully nonlinear income tax and obtain a carbon wedge equal to the social cost of carbon at every history under weak separability and any preference satisfying mild conditions, alongside a Mirrleesian labor wedge and a non-zero savings wedge following the inverse Euler condition. The two approaches converge in their policy conclusions on the condition that the tax system is jointly optimized. What matters is therefore not so much linearity of instruments, but whether other instruments are misset.

3 Environment

The economy is a two-good extension of the dynamic Mirrlees setting of Golosov et al. (2003), Farhi and Werning (2013), and Golosov et al. (2016). The two goods are a clean numéraire and a polluting good. To isolate how household income inequality shapes the optimal carbon tax, I abstract from international and intergenerational equity and consider a single country as well as a single generation living for $T + 1$ periods. With a single generation, calendar time and age are equivalent, so a time-varying tax is age-varying.

Each household consumes a clean good c_t and a polluting good q_t , supplies labor l_t , and is exposed to the aggregate pollution stock S_t . Expected lifetime utility is

$$\mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U(c_t, q_t, l_t, S_t) \right]$$

$\beta \in (0, 1)$ is the discount factor and utility is non-separable between its arguments. The clean good c is the numéraire, with prices normalized to 1. The dirty good is assumed to be priced at 1 as well. The clean good is emissions-free, whereas each unit of the dirty good emits a unit of carbon, so that date- t emissions equal aggregate dirty-good consumption Q_t . Emissions contribute to a persistent pollution stock S_t , a function of current and past aggregate emissions.

$$S_t = S(Q_0, Q_1, \dots, Q_t)$$

$Q_t = \int_0^\infty \mathbb{E}_0[q_t(\theta^t)|\theta] dF_0(\theta)$ is aggregate consumption of the dirty good at date t . I assume that S is differentiable and non-decreasing in each argument, but otherwise leave it unrestricted. The linear special case $S_t = \sum_{j=0}^t (1 - d_j) Q_{t-j}$ in which each past emission flow leaves $(1 - d_j)$ in the current stock is the carbon-cycle representation of Golosov et al. (2014). The general

form above nests it and allows nonlinear accumulation. A convenient leading example keeps a fraction δ of the stock from one period to the next.

$$S_t = \sum_{j=0}^t \delta^j Q_{t-j}$$

This is the recursion $S_t = \delta S_{t-1} + Q_t$ from an initial pollution stock. To keep the state finite, I assume emissions stop affecting the pollution stock after s periods.

$$S_t = S(Q_{t-s}, \dots, Q_t)$$

Aside from the polluting good, the model follows the standard dynamic Mirrlees setup. At $t = 0$, agents draw an initial skill (type) θ_0 from a distribution $F_0(\theta)$. Later skills follow a Markov process $F_t(\theta|\theta_{t-1})$. At period t , each agent knows their skill draws up to time t , which I denote by a skill history vector $\theta^t = (\theta_0, \dots, \theta_t)$. Θ^t is the set of all possible skill histories.

Agents retire at an exogenous date $T_R \leq T$. From T_R on, the agent supplies no labor and draws no further skills. At that point, there is no more screening problem. However, the agent continues to save and consume. This is the life-cycle structure of Farhi and Werning (2013), aside from the presence of the dirty good.

An agent of type θ_t who supplies l_t units of labor produces $y_t = \theta_t l_t$ units of output. Everyone observes total output, but skill θ_t and labor supply l_t are private. Consumption of both goods is also publicly observable¹. Let $c_t(\theta^t), q_t(\theta^t) : \Theta^t \rightarrow \mathbb{R}_+$ and $y_t(\theta^t) : \Theta^t \rightarrow \mathbb{R}_+$ be the agent's allocation of the clean good, the polluting good, and output, respectively. Let $\sigma^t(\theta^t) : \Theta^t \rightarrow \Theta^t$ describe how an agent with skill history θ^t reports it, and let Σ^t be the set of all possible reporting strategies. Capital transfers resources across periods through a linear technology with exogenous constant gross return $R > 0$. The social welfare function is

$$\int_0^\infty \alpha(\theta) \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U(c_t, q_t, l_t, S_t) \right] dF_0(\theta)$$

$\alpha(\theta)$ is a Pareto weight on the welfare of an agent with initial skill θ , normalized so that $\int_0^\infty \alpha(\theta) dF_0(\theta) = 1$. A utilitarian planner sets $\alpha(\theta) = 1$ for all θ . Let U_c and U_q denote the partial derivatives with respect to the clean and polluting goods, and U_l with respect to labor.

¹This is a plausible assumption for some goods such as electricity and air travel. For goods whose consumption cannot be as easily observed, the planner is further constrained and the history-dependent commodity wedges of Sections 4 and 5.1 no longer apply. However, the results with an anonymous price of Section 5.2 still apply.

Since $y = \theta l$, I can write $U(c, q, l, S)$ as $U(c, q, \frac{y}{\theta}, S)$. Let U_y and U_θ be the partial derivatives of utility with respect to y and θ .

Assumption 3.1 (Preferences).

U is twice continuously differentiable with $U_c, U_q > 0$, $U_l, U_S < 0$, $U_{cc}, U_{qq}, U_{ll}, U_{SS} \leq 0$, strictly quasi-concave in (c, q) with $2U_c U_q U_{cq} - U_c^2 U_{qq} - U_q^2 U_{cc} > 0$ (regular strict quasi-concavity: a strictly convex, non-degenerate indifference map), and satisfies Inada conditions in (c, q) . Single crossing holds at every S .

$$\frac{\partial}{\partial \theta} \frac{U_l(c, q, y/\theta)}{U_c(c, q, y/\theta, S)} \geq 0 \quad \text{for all } (c, q, y, \theta)$$

At the optimal allocation, $1 + \epsilon_t(\theta) - \gamma_{c,t}(\theta) > 0$, where $\epsilon_t, \gamma_{c,t}$ are defined in (1) below. The conditions involving labor are only imposed on working periods, since no work is supplied after retirement.

Assumption 3.2 (Shock process).

For all $1 \leq t < T_R$, f_t is differentiable in both arguments with $f'_t = \partial f_t / \partial \theta$ and $f_{2,t} = \partial f_t / \partial \theta_{t-1}$. For all θ_{t-1} , the ratio

$$\frac{\theta_{t-1} \int_{\theta}^{\infty} f_{2,t}(x|\theta_{t-1}) dx}{\theta f_t(\theta|\theta_{t-1})}$$

is bounded, and $\lim_{\theta \rightarrow \infty} \frac{1 - F_t(\theta|\theta_{t-1})}{\theta f_t(\theta|\theta_{t-1})}$ is finite.

Any admissible allocation must be incentive compatible, meaning the agent must weakly prefer truthful reporting to any other strategy. The pollution stock S_t is an aggregate, so no single agent's deviation moves it and it enters both sides of the constraint identically. For every initial type θ and every reporting strategy,

$$\begin{aligned} & \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) \middle| \theta \right] \\ & \geq \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\sigma^t(\theta^t)), q_t(\sigma^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{\theta_t}, S_t \right) \middle| \theta \right] \end{aligned}$$

Over a finite horizon, global incentive compatibility is equivalent to the absence of a profitable one-shot deviation after every history (Fernandes and Phelan 2000; Kapička 2013). I follow the first-order approach of Kapička (2013), Farhi and Werning (2013), and Golosov et al. (2016), replacing these constraints with their local counterpart. Let $u_t(\theta^t)$ be the lifetime utility an

agent obtains from reporting truthfully at every date. Truth-telling is locally optimal when the envelope condition holds.

$$\dot{u}_t(\theta^t) = U_\theta \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) + \beta w_{2,t}(\theta^t)$$

$w_{2,t}(\theta^t) = \int_0^\infty u_{t+1}(\theta^t, \theta_{t+1}) \partial_{\theta_t} f_{t+1}(\theta_{t+1} | \theta_t) d\theta_{t+1}$ is the marginal continuation value of type θ^t , and the dot denotes a derivative with respect to the current draw θ_t . The continuation term is the future information rent of the current draw. Under Assumptions 3.1 and 3.2, this local condition is necessary for incentive compatibility (Kapička 2013). The local conditions define a relaxed problem and are necessary but not sufficient for global incentive compatibility. Indeed, no general primitive restriction guarantees sufficiency in dynamic models with multiple goods (Goloso et al. 2016). I solve the relaxed problem throughout and validate global incentive compatibility in the numerical exercise of Section 6.

The aggregate resource constraint must hold in every period,

$$\int_0^\infty \mathbb{E}_0[c_t(\theta^t) + q_t(\theta^t) | \theta] dF_0(\theta) + K_{t+1} \leq \int_0^\infty \mathbb{E}_0[y_t(\theta^t) | \theta] dF_0(\theta) + RK_t$$

Here, K_t is the aggregate capital stock, with K_0 given and $K_{T+1} = 0$. Dividing each period's constraint by R^t , summing over t , and using these boundary conditions to telescope the capital terms consolidates the sequence into a single intertemporal constraint,

$$\int_0^\infty \mathbb{E}_0 \left[\sum_{t=0}^T \frac{c_t(\theta^t) + q_t(\theta^t)}{R^t} \middle| \theta \right] dF_0(\theta) \leq \int_0^\infty \mathbb{E}_0 \left[\sum_{t=0}^T \frac{y_t(\theta^t)}{R^t} \middle| \theta \right] dF_0(\theta) + RK_0$$

The planner chooses allocations to maximize welfare subject to the local incentive constraint, the intertemporal resource constraint, and the pollutant law of motion.

$$\max_{\{c_t(\theta^t), q_t(\theta^t), y_t(\theta^t), S_t\}_{\theta^t \in \Theta^t, t=0, \dots, T}} \int_0^\infty \alpha(\theta) \left(\mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) \middle| \theta \right] \right) dF_0(\theta)$$

Three wedges, the implicit marginal taxes on the labor, savings, and commodity margins, describe the distortions the optimal allocation imposes.

$$\begin{aligned}
1 - \tau_t^y(\theta^t) &\equiv -\frac{U_l\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}{\theta_t U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)} \\
1 - \tau_t^s(\theta^t) &\equiv \frac{1}{\beta R} \frac{U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}{\mathbb{E}_t\left[U_c\left(c_{t+1}(\theta^{t+1}), q_{t+1}(\theta^{t+1}), \frac{y_{t+1}(\theta^{t+1})}{\theta_{t+1}}, S_{t+1}\right)\right]} \\
1 + \tau_t^q(\theta^t) &\equiv \frac{U_q\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}{U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}
\end{aligned}$$

I define seven preference statistics. The first two measure the own curvature of labor disutility and clean-good utility, holding the other arguments fixed.

$$\epsilon_t \equiv \frac{U_{ll,t} l_t}{U_{l,t}} \quad \sigma_t \equiv -\frac{U_{cc,t} c_t}{U_{c,t}} \quad (1)$$

Five are cross-partials that measure how the marginal utilities of the two goods and of the pollutant move with labor as well as with each other. In order, they represent the labor complementarity with the clean good, the labor complementarity with the dirty good, the labor complementarity of environmental quality, the cross-elasticity between the two goods, and the complementarity of the pollutant with the clean good.

$$\gamma_{c,t} \equiv \frac{U_{cl,t} l_t}{U_{c,t}} \quad \gamma_{q,t} \equiv \frac{U_{ql,t} l_t}{U_{q,t}} \quad \xi_t \equiv \frac{U_{Sl,t} l_t}{U_{S,t}} \quad \sigma_{cq,t} \equiv -\frac{U_{cq,t} q_t}{U_{c,t}} \quad \xi_{c,t} \equiv \frac{U_{Sc,t} c_t}{U_{S,t}} \quad (2)$$

Note that complementarity is measured with respect to labor rather than leisure here, so a positive γ signals a good whose marginal utility rises as the agent works more. To get the leisure complementarity interpretation, it suffices to flip the sign. Most of the commodity-wedge results hinge on the gap $\gamma_{c,t} - \gamma_{q,t}$ between the clean and dirty goods' complementarities with labor. If the gap is positive, the clean good is the relative labor complement, meaning an agent who works more wants more of it. Likewise, the results pertaining to the social cost of carbon hinge on the gap $\xi_t - \gamma_{c,t}$, which measures how the marginal damage of pollution varies with labor supply. If the gap is positive, an agent who works more suffers more from pollution and a cleaner environment is a relative labor complement. Both complementarities are proportional to l_t , so both disappear in retirement.

Two willingness-to-pay objects also describe the result. $U_{q,t}/U_{c,t}$ is what the household would pay for a unit of the dirty good and $-U_{S,t}/U_{c,t}$ what it would pay for a cleaner environment. For $z \in \{c, y, q\}$,

$$\mathcal{I}_{z,t}^q \equiv \frac{\partial}{\partial z_t} \ln \left(\frac{U_{q,t}}{U_{c,t}} \right) \quad \mathcal{I}_{z,t}^S \equiv \frac{\partial}{\partial z_t} \ln \left(\frac{-U_{S,t}}{U_{c,t}} \right) \quad (3)$$

From here, I define four statistics with respect to the numéraire and labor earnings.

$$\begin{aligned} \mathcal{I}_{c,t}^q &= \frac{\sigma_t}{c_t} - \frac{\sigma_{cq,t}}{q_t(1 + \tau_t^q)} & \mathcal{I}_{y,t}^q &= -\frac{\gamma_{c,t} - \gamma_{q,t}}{y_t} \\ \mathcal{I}_{c,t}^S &= \frac{\sigma_t + \xi_{c,t}}{c_t} & \mathcal{I}_{y,t}^S &= \frac{\xi_t - \gamma_{c,t}}{y_t} \end{aligned} \quad (4)$$

$\mathcal{I}_{c,t}^q$ is the rate at which willingness to pay for the dirty good rises with clean-good consumption. A positive value means an agent with more to spend buys more of the dirty good, so $\mathcal{I}_{c,t}^q > 0$ holds if and only if that good is normal (Appendix C.1). $\mathcal{I}_{c,t}^S$ is positive where more consumption raises the value of a cleaner environment. Its sign carries no normality reading however since the household does not buy environmental quality. The labor earnings gradients say the same about work and are equivalent to the aforementioned complementarity gaps. Indeed, $\mathcal{I}_{y,t}^q < 0$ also means that the clean good is the relative labor complement, so the household would pay less for a unit of the dirty good the more it works.

4 Full optimum

4.1 Planner's problem

First, following Kapička (2013)², we can write the planner's problem recursively. Appendix A.2 shows that the presence of externalities does not invalidate the approach. No individual report moves the aggregate pollution stock, so the agent's incentive problem is unchanged, aside from the pollution stock entering utility as a date-specific parameter. Aggregates are deterministic, so internalizing them appends a single new state, the emission window $\mathbf{S}_- = (Q_{t-s}, \dots, Q_{t-1})$ of lagged aggregate emissions feeding current and future pollution. The continuation value receives the updated window with the current aggregate Q_t appended.

$$\begin{aligned}
 V_t(\hat{w}, \hat{w}_2, \theta_-, \mathbf{S}_-) &= \min_{c, q, y, u, w, w_2, S} \int_0^\infty \left[c(\theta) + q(\theta) - y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta, \mathbf{S}) \right] f_t(\theta | \theta_-) d\theta \\
 &\quad \text{s.t.} \\
 \dot{u}(\theta) &= U_\theta \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right) + \beta w_2(\theta) \\
 \hat{w} &= \int_0^\infty u(\theta) \bar{\alpha}_t(\theta) f_t(\theta | \theta_-) d\theta \\
 \hat{w}_2 &= \int_0^\infty u(\theta) f_{2,t}(\theta | \theta_-) d\theta \\
 u(\theta) &= U \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right) + \beta w(\theta) \\
 S &= S \left(\dots, \int q(\theta) f(\theta) d\theta \right)
 \end{aligned}$$

The planner minimizes resource cost while offering an agent expected lifetime utility $\hat{w}$ with marginal continuation value $\hat{w}_2$, now internalizing what the assigned dirty good does to the pollution stock. $\mathbf{S}$ is the updated emission state, $\mathbf{S}_-$ with Q appended. $\bar{\alpha}_t(\theta)$ is the continuation Pareto weight attached to delivering utility to type θ . It equals the primitive Pareto weight in the initial period, $\bar{\alpha}_0(\theta) = \alpha(\theta)$, and $\bar{\alpha}_t(\theta) = 1$ for $t > 0$. As in Golosov et al. (2016), the lifetime Pareto weights enter only at date 0. From date 1 on, the weight collapses to one.

²Kapicka's consumption set is convex and compact (Assumption 1), so consumption can be a vector of two or more goods. Therefore, his results carry over and are applicable in my setting.

4.2 Optimal distortions

Solving the planner's problem delivers an allocation respecting the following three wedges in terms of labor supply, savings, and the split between the two goods.

Proposition 4.1 (Optimal wedges).

The labor wedge is

$$\frac{\tau_t^y}{1 - \tau_t^y} = A_t(\theta)B_t(\theta)C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} D_t(\theta)$$

$$A_t(\theta) = 1 + \epsilon_t(\theta) - \gamma_{c,t}(\theta)$$

$$B_t(\theta) = \frac{1 - F_t(\theta)}{\theta f_t(\theta)}$$

$$C_t(\theta) = \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x}) \frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_{c,t}(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] (1 - \lambda_{1,t} \bar{\alpha}_t(x) U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)}$$

$$D_t(\theta) = \frac{A_t(\theta)}{A_{t-1}} \frac{U_{c,t}(\theta)}{U_{c,t-1}} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta f_t(\theta)}$$

Let $\widehat{U}_{c,t} \equiv \frac{U_{c,t}}{1 - \chi_t}$, with $\chi_t \equiv \frac{\gamma_{c,t}}{1 + \epsilon_t - \gamma_{c,t}} \frac{\tau_t^y}{1 - \tau_t^y}$. Savings satisfy the generalized inverse Euler condition

$$\frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1}{\widehat{U}_{c,t+1}} \right]$$

The carbon wedge departs from the social cost of carbon by

$$\frac{\tau_t^q - \text{SCC}_t}{1 + \tau_t^q} = \frac{\gamma_{c,t} - \gamma_{q,t}}{1 + \epsilon_t - \gamma_{c,t}} \frac{\tau_t^y}{1 - \tau_t^y} \quad (5)$$

with the social cost of carbon SCC_t characterized in Proposition 4.2.

Proof. See Appendices A.3, A.4, and A.5. □

Corollary 4.1 (Separable preferences).

Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$, the corrective and redistributive margins separate.

$$\tau_t^q = \text{SCC}_t \quad \frac{\tau_t^y}{1 - \tau_t^y} = A_t B_t C_t + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} D_t \quad \frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1}{\widehat{U}_{c,t+1}} \right]$$

Proof. Immediate from (5) at $\gamma_{c,t} = \gamma_{q,t}$. □

The first formula is the standard ABCD labor wedge formula, as in Golosov et al. (2016) and Farhi and Werning (2013). None of the terms change, except for term C_t which represents the social value of the resources extracted from types above θ , where the exponential term captures the income effect the redistribution induces on those types. This term now includes the cross-curvature between the two goods, such that the wedge rises all things equal when the goods are Edgeworth substitutes ($\sigma_{cq,t} > 0$) and falls when they are complements ($\sigma_{cq,t} < 0$). Intuitively, when the goods are substitutes, taking away the numéraire from these types renders the dirty good more pleasurable and allows the planner to extract more resources from them.

The second formula is the generalized inverse Euler condition, expressed as a reweighted inverse Euler condition similar to Hellwig (2021). When preferences are non-separable between labor and consumption, the cost of delivering utility is weighted by $1 - \chi_t$, where χ_t is the incentive credit. When the numéraire complements labor, a unit of consumption makes work less painful and shrinks the information rent that truth-telling requires, rebating χ_t from the delivery cost. Neither the second good nor the externality changes the structure of savings wedges.

The third formula describes how the planner allocates each good across types. It is the dynamic analogue to the optimal commodity wedge derived in Jacobs and Mooij (2015). The left-hand side is the commodity distortion net of the social cost of carbon described in the next section and the right-hand side is the labor distortion scaled by the relative labor complementarities of the two goods. This labor complementarity gap has a willingness to pay gradient reading, equal to $-y_t \mathcal{I}_{y,t}^q$ by (4), so a positive gap says a household supplying more labor would pay less for a unit of the dirty good. When preferences are weakly separable, the carbon wedge is set at the social cost of carbon. With no externalities, we get the dynamic Atkinson-Stiglitz result of Golosov et al. (2003). Otherwise, if the numéraire is the relative labor complement or, read equivalently, if the household's willingness to pay for the dirty good falls with labor supply, taxing the dirty good is an indirect tax on leisure. The carbon wedge is then above the social cost of carbon and inherits both the shape and the life-cycle path of the labor wedge, adjusted for the ratio of relative labor complementarities and labor response elasticities. Compared to the static commodity wedge of Mirrlees (1976) or its environmental extension in Jacobs and Mooij (2015), the dynamic commodity wedge is a dynamic object. Therefore, it should optimally not only vary across types, but also across age.

4.3 Social cost of carbon

The social cost of carbon in (5) is the discounted stream of marginal damages, each term adjusted for how differently pollution and the numéraire bear on labor supply.

Proposition 4.2 (The social cost of carbon).

The shadow price of current emissions (13) is

$$\text{SCC}_t = - \sum_{j=0}^s \frac{1}{R^j} \mathbb{E}_0 \left[\left(1 + \frac{\xi_{t+j} - \gamma_{c,t+j}}{1 + \epsilon_{t+j} - \gamma_{c,t+j}} \frac{\tau_{t+j}^y}{1 - \tau_{t+j}^y} \right) \frac{U_{S,t+j}}{U_{c,t+j}} \Phi_j \right] \quad (6)$$

with $\Phi_j \equiv \partial S_{t+j} / \partial Q_t$ the effect of current aggregate emissions on the date- $(t+j)$ pollution stock ($\Phi_j = 0$ for $j > s$ by construction). The expectation $\mathbb{E}_0$ averages the full date- $(t+j)$ population of skill histories. With deterministic aggregates, the social cost of carbon is common across households at each date. We get the Pigouvian price of carbon when the factor is one.

$$\text{SCC}_t^P = - \sum_{j=0}^s \frac{1}{R^j} \mathbb{E}_0 \left[\frac{U_{S,t+j}}{U_{c,t+j}} \Phi_j \right] \quad (7)$$

$\text{SCC}_t = \text{SCC}_t^P$ when $\xi_{t+j} = \gamma_{c,t+j}$ at every future history or when labor is undistorted.

Proof. See Appendix A.6. □

The bracket is a fiscal-adjustment factor, active wherever labor is distorted and pollution and the numéraire interact with labor differently. The gap has the same willingness-to-pay reading as the commodity wedge's. By (4), $-U_{S,t+j}/U_{c,t+j}$ is what the household would pay for a cleaner environment in units of the numéraire, and $\xi_{t+j} - \gamma_{c,t+j} = y_{t+j} \mathcal{I}_{y,t+j}^S$ measures how that willingness to pay moves with work. When it rises with work, pollution hits agents twice, directly through utility and again by discouraging work, making the social cost of carbon exceed the Pigouvian level. When it falls with work, the second channel runs the other direction and the social cost of carbon is under the Pigouvian level. The correction applies to every term of the discounted stream and is the dynamic extension of Jacobs and Mooij (2015).

5 Constrained optimum

The results so far let the government reach any incentive-compatible allocation, but real carbon taxes may be set inside imperfect tax systems where the income tax is not the one the dynamic Mirrlees optimum would prescribe. To model this restriction, I consider the planner problem in which labor and savings wedges are held at exogenous schedules. That is, of the incentive-feasible allocations, the planner may only offer those producing the required wedges.

This question has some antecedents in the Ramsey literature. Barrage (2020) computes optimal carbon taxes in a representative-agent Ramsey model with either the labor or the capital income tax held at its observed level, whereas Douenne et al. (2026) extend her framework to heterogeneous agents. Douenne et al. (2025) further extend the model into an incomplete markets setting and compute optimal carbon taxes with both income taxes fixed. These analyses only consider linear tax instruments as well as linear tax restrictions, whereas this paper allows for nonlinear tax instruments and nonlinear tax restrictions.

Keeping income wedges fixed rules out using carbon revenue to cut marginal income tax rates, as in the revenue-neutral swaps of British Columbia and Sweden, or to fund income-contingent transfers such as France's Chèque énergie. Since the planner chooses the best possible allocation under these restrictions, it implicitly recycles any carbon revenue optimally. The theory does not offer a revenue recycling rule, but the quantitative results of Section 6 suggest that implicit recycling is optimally close to a uniform lifetime dividend, consistent with the fee-and-dividend design of Canada's now-repealed federal fuel charge and Switzerland's CO₂ levy.

5.1 Given income tax

Let the labor wedge be held at a schedule $\bar{\tau}_t^y(\theta) < 1$ with Lagrange multiplier $\eta_t(\theta)f_t(\theta|\theta_-)$.

$$\Gamma_t(\theta) \equiv -\frac{U_l\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right)}{\theta U_c\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right)} - (1 - \bar{\tau}_t^y(\theta)) = 0 \quad (8)$$

$\Gamma_t(\theta)$ is the distance between the implemented labor wedge and the fixed schedule. Moreover, the savings wedge is held at $\bar{\tau}_t^s(\theta_t) < 1$ with Lagrange multiplier $\vartheta_{t+1}(\theta^t)f_t(\theta|\theta_-)$.

$$U_{c,t}(\theta^t) = (1 - \bar{\tau}_t^s(\theta_t)) \beta R \mathbb{E}_t[U_{c,t+1}(\theta^{t+1})] \quad (9)$$

Because the savings-wedge constraint involves current and future consumption, the date- t problem cannot enforce it alone. Given a marginal utility of current consumption, the constraint requires tomorrow's marginal utility to average some value M . Today's consumption

is likewise constrained by the requirement $\hat{M}$ inherited from yesterday's problem. This new state is needed since two continuation plans delivering the same promised utility and marginal continuation value can still differ in their expected future marginal utility, even if the savings-wedge constraint admits only one of them. The planner's problem is that of Section 4 with wedge constraints (8) and (9), M in the continuation value, and state $\hat{M}$.

Lemma 5.1 (Recursive formulation with fixed wedges).

Neither constraint involves a reporting strategy or enters incentive constraints, so the first-order approach still applies. The savings-wedge constraint reaches the date- $(t + 1)$ allocations only through $\hat{M}$, which enters as a new state with multiplier $-\vartheta_t$, such that $\vartheta_0 = 0$ and $\vartheta_{T+1} = 0$.

Proof. See Appendix B.1. □

As shown in Appendix B.1, the labor-wedge constraint adds a term to every first-order condition, while the savings-wedge constraint adds two. The first comes from $\hat{M}$ inherited from the date- $(t - 1)$ problem, and the second from M passed forward to the date- $(t + 1)$ problem. These two terms can be collected into a single shadow price of the intertemporal constraint:

$$\eta_t^s(\theta^t) \equiv \left[\vartheta_t - \frac{\vartheta_{t+1}(\theta^t)}{(1 - \bar{\tau}_t^s) \beta R^2} \right] U_{c,t}(\theta^t) \quad (10)$$

Hence, η_t prices the labor-wedge constraint and η_t^s prices the savings-wedge constraint. Both enter the first-order conditions through $\zeta_t \equiv \eta_t(1 - \bar{\tau}_t^y) - \eta_t^s$ and η_t enters once more on its own.

Proposition 5.1 (Constrained-optimal carbon price).

With the willingness-to-pay responses $\mathcal{I}_{c,t}^q$ and $\mathcal{I}_{y,t}^q$ of (3) and $\tau_t^{y,*}$ the desired labor wedge,

$$\frac{\tau_t^q - \text{SCC}_t}{1 + \tau_t^q} = \frac{\gamma_{c,t} - \gamma_{q,t}}{1 + \epsilon_t - \gamma_{c,t}} \frac{\tau_t^{y,*}}{1 - \tau_t^{y,*}} - \eta_t \mathcal{I}_{y,t}^q - \zeta_t \mathcal{I}_{c,t}^q \quad (11)$$

The social cost of carbon keeps the usual form, with the fiscal adjustment term replaced by

$$1 + \frac{\xi_{t+j} - \gamma_{c,t+j}}{1 + \epsilon_{t+j} - \gamma_{c,t+j}} \frac{\tau_{t+j}^{y,*}}{1 - \tau_{t+j}^{y,*}} + \eta_{t+j} \mathcal{I}_{y,t+j}^S + \zeta_{t+j} \mathcal{I}_{c,t+j}^S$$

Proof. See Appendices B.1 and B.4, with Appendix A.7 for $\psi = 0$ at the retirement dates. □

The first term of the commodity wedge has the same form as that in the full optimum, with the labor wedge being the one the planner wants but cannot necessarily achieve.

The second term reflects the shadow cost of the labor-wedge constraint. A positive η_t means that lowering the labor wedge would reduce the planner's resource cost of delivering an allocation; the planner would like to lower the labor wedge but cannot. For brevity, I will call this "overdistortion". Likewise, I will call the opposite case "underdistortion". $\mathcal{I}_{y,t}^q$ measures how willingness to pay for the dirty good changes with labor income. Suppose $\eta_t > 0$ and the dirty good is the relative labor substitute, $\mathcal{I}_{y,t}^q < 0$. Then, $-\eta_t \mathcal{I}_{y,t}^q > 0$, pulling up the carbon wedge. Intuitively, shifting expenditure from the dirty good to the clean good makes work less painful, so a higher carbon price encourages work and undoes part of the labor distortion.

The third term captures how the consumption mix affects the two wedge constraints through the marginal utility of the clean good. ζ_t is the net shadow price of a proportional change in this marginal utility, combining the labor-wedge and savings-wedge constraint multipliers. $\mathcal{I}_{c,t}^q$ measures how willingness to pay for the dirty good changes with clean-good consumption and is positive when the dirty good is normal. Set the savings multiplier aside and suppose $\eta_t > 0$, so that $\zeta_t = \eta_t(1 - \bar{\tau}_t^y) > 0$. If the dirty good is normal, $\mathcal{I}_{c,t}^q > 0$, then $-\zeta_t \mathcal{I}_{c,t}^q < 0$, pulling down the carbon wedge. Intuitively, shifting expenditure from the clean good to the dirty good raises the marginal utility of the clean good, strengthening the incentive to work. This channel counteracts the previous one here, so the net effect of the terms depends on their relative size. Adding back η_t^s to ζ_t , $\eta_t^s < 0$ reinforces the pull, while $\eta_t^s > 0$ weakens or even reverses it.

The fiscal adjustment term to the social cost of carbon also gains two terms. If $\eta_{t+j} > 0$ and having a higher labor income raises the willingness to pay for environmental quality, $\mathcal{I}_{y,t+j}^S > 0$, then the term is positive, pulling up the social cost of carbon. If we set aside the savings multiplier η_t^s in ζ_t such that $\zeta_t > 0$ and if consuming more of the clean good raises the willingness to pay for environmental quality, $\mathcal{I}_{c,t+j}^S > 0$, then the term is once again positive, further pulling up the social cost of carbon. Hence, the planner wants to encourage work and can do so by providing more environmental quality, which high earners and high spenders value more.

Corollary 5.1 (Constrained-optimal carbon price with separable preferences).

Under weak separability or $\gamma_{c,t} = \gamma_{q,t}$,

$$\frac{\tau_t^q - \text{SCC}_t}{1 + \tau_t^q} = -\zeta_t \mathcal{I}_{c,t}^q$$

When the dirty good is normal, the carbon price is below (above) it where $\zeta_t > 0$ ($\zeta_t < 0$).

Proof. Proposition 5.1 at $\gamma_{c,t} = \gamma_{q,t}$. □

If labor or savings wedges are fixed, then weak separability no longer ensures that the carbon price is set at the social cost of carbon. When consumption moves the willingness to pay for the dirty good, the carbon price can be used to offset the constraints.

5.1.1 Savings-wedge constraint over the life cycle

Whereas the sign of the labor-wedge multiplier is determined by whether labor is underdistorted or overdistrorted at a given date and history, acting as a local "push", that of the savings-wedge multiplier pulls in opposite directions at different dates, acting as a "pivot" across the life cycle.

Lemma 5.2 (Sign of the savings multiplier).

Along any history on which η_t^s is nonzero at some date, it must take both positive and negative values over the life cycle. The result holds for any labor schedule and skill process.

Proof. See Appendix B.3. □

Intuitively, consider the case where savings is always overdistrorted. If we were talking about the labor wedge, this would only affect consumption-labor choices within a single date. However, the savings wedge here frontloads consumption. Because savings is made to offer a lower return, the household consumes more today and less tomorrow. Therefore, the planner would like to reduce consumption today and increase it tomorrow. Without the means to directly affect the savings wedge, the planner's second option is to affect the consumption mix, if consumption of one good affects demand for the other good. That is, if the dirty good encourages consumption of the clean good, then the planner counteracts the frontloading by distorting the dirty good more today and less tomorrow, shifting clean good consumption from today to tomorrow.

This lemma shows that both signs must appear, but does not specify their order or timing. Under the additional curvature and positivity conditions in Appendix B.3, a deterministic skill path with a fixed savings wedge strictly above the preferred wedge at every transition yields a positive η_t^s early and a negative one late, with a single sign change. With stochastic skills, the order and timing depend on the solved allocation.

In retirement, the labor and screening terms disappear, so equation (11) reduces to

$$\frac{\tau_t^q - \text{SCC}_t}{1 + \tau_t^q} = \eta_t^s \mathcal{I}_{c,t}^q$$

For a normal dirty good, $\mathcal{I}_{c,t}^q > 0$, the retirement carbon-price correction has the sign of η_t^s . The same savings term appears during working life without weak separability. At $t < T_R$, expanding ζ_t in Proposition 5.1 gives an equivalent expression.

$$\frac{\tau_t^q - \text{SCC}_t}{1 + \tau_t^q} = \frac{\gamma_{c,t} - \gamma_{q,t}}{1 + \epsilon_t - \gamma_{c,t}} \frac{\tau_t^{y,*}}{1 - \tau_t^{y,*}} - \eta_t \left[\mathcal{I}_{y,t}^q + (1 - \bar{\tau}_t^y) \mathcal{I}_{c,t}^q \right] + \eta_t^s \mathcal{I}_{c,t}^q$$

The "pivot" effect is strengthened or weakened by the "push" effect, depending on their signs.

5.1.2 Carbon-only reforms and implementation

Because actual governments do not actually condition taxes on what they "think" one's skill is nor on what households "report" their skill to be but rather on observables such as income or, perhaps, consumption, we need to translate the previous results. First, suppose the labor-wedge constraint corresponds to a labor earnings function $T_t^y(y)$ and the capital-wedge constraint is delivered by a fixed capital income tax rate of τ^k ³. That is, a labor income tax does not depend on assets and the capital income tax do not depend on labor income. Likewise, the carbon wedge is implemented by a tax on dirty-good consumption that may depend on dirty-good consumption, but not on the other observables. This allows to cleanly hold the rest of the tax system fixed while varying the carbon price. Without this restriction, labor or savings wedges, supposed to be fixed, could be implemented with a sufficiently cleverly-designed carbon tax. For example, with $\tau^y = T^y'(y) + qt'(y)$, the labor-conditioned carbon tax side can step in to do the work of the labor tax. For a discussion of the distinction between wedges and implementation, including how various instruments can implement the same allocation, see Golosov et al. (2007).

These conditions do not necessarily establish that the allocation can be implemented by such a tax. A household deviating in the direct mechanism receives the entire package assigned to another reporting history. Under a tax on purchase histories, it can instead choose another purchase history while re-optimizing hours and saving. Global incentive compatibility of the direct mechanism therefore need not rule out profitable deviations under the tax. Appendix B.5 gives a counterexample for the candidate tax construction. Verifying implementation of the computed allocation requires checking that no household gains by changing purchases, hours, or saving under the proposed tax system, following the ex post verification logic of Ábrahám and Pavoni (2008).

The numerical comparisons in Section 6 measure the welfare value of carbon-wedge flexibility across computed relaxed allocations, with consumption levels and implied transfers re-optimized in each regime. This includes the linear-rate comparisons, since fixing the marginal carbon price does not fix the transfers. These comparisons do not by themselves establish welfare gains from implemented reforms. The progressive numerical benchmark also uses normalized earnings y_t/θ_{t-1} and therefore does not impose a common tax schedule on observed earnings: households with equal earnings at the same date can face different fixed labor wedges (Appendix D.1).

5.2 Optimal common carbon price

The optimal common carbon price averages the desired corrections to the social cost of carbon, with greater weight on households whose dirty-good demand responds more to a compensated

³The current quantitative exercise assumes a single capital income tax rate. An extension will seek to calibrate the capital income tax schedule in income and age.

price change. With income wedges free, these corrections reflect screening. Constraining the income wedges adds the labor and savings corrections derived above. Holding the rate constant over the life cycle requires balancing these corrections across skills and dates.

First, leave the income wedges free and restrict the relaxed allocation problem to a consumer price $\bar{p}_t = 1 + \bar{\tau}_t^q > 0$ common across all date- t histories. The constraint is $U_q/U_c = \bar{p}_t$. Every household faces the same marginal price at a given date, but the planner can vary that price across dates. Let $s_{qq,t} < 0$ denote the compensated response of dirty-good demand to its price, holding labor and pollution fixed, and define

$$\varpi_t \equiv \frac{\bar{\tau}_t^q - \text{SCC}_t}{\bar{p}_t} \quad \text{and} \quad \varpi_t^* \equiv (\gamma_{c,t} - \gamma_{q,t}) \frac{\psi U_{c,t}}{\theta f_t}$$

ϖ_t measures the carbon wedge's deviation from the social cost of carbon as a share of the consumer price. The history-specific term ϖ_t^* is the screening correction from Section 4, evaluated at the common-price allocation.

Proposition 5.2 (The optimal common price).

With income wedges free, an interior optimal date-specific common price satisfies

$$\varpi_t = \frac{\mathbb{E}_0[|s_{qq,t}| \varpi_t^*]}{\mathbb{E}_0[|s_{qq,t}|]}$$

If consumption is weakly separable from labor at every date, $\varpi_t^* = 0$ at every history. Then $\bar{\tau}_t^q = \text{SCC}_t$, and the common-price restriction entails no loss in the relaxed allocation problem of Corollary 4.1. In retirement, $\psi = 0$, so the optimal date-specific rate equals the social cost of carbon even without weak separability during working life.

Proof. See Appendices C.2 and C.4, and Appendix A.7 for $\psi = 0$. □

The weights measure how much a marginal price change moves dirty consumption at constant utility. When the desired correction exceeds the common correction, raising the price contributes to reducing the planner's resource cost. When it falls short, the contribution has the opposite sign. The optimum balances these contributions across households.⁴

This average is an implicit optimality condition. The demand responses, screening corrections, and social cost of carbon all belong to the common-price allocation, so the formula does not average carbon wedges from a separately solved unrestricted optimum. The social cost of carbon is also the shadow emission cost of this constrained problem: the price restriction can change

⁴The role of compensated substitution responses parallels the many-person Ramsey rule of Diamond (1975). Here, the redistributive motives are summarized by the desired corrections being averaged.

the valuation of damages as well as aggregate emissions. Appendix C.4 derives the additional term in the damage weights.

Constraining the income wedges preserves the averaging rule but changes the corrections being averaged. Adding the terms from Proposition 5.1 gives the desired correction

$$\varpi_t^{**} \equiv \begin{cases} \varpi_t^* + (\gamma_{c,t} - \gamma_{q,t}) \frac{\eta_t}{y_t} - \zeta_t \mathcal{I}_{c,t}^q, & t < T_R, \\ \eta_t^s \mathcal{I}_{c,t}^q, & t \geq T_R. \end{cases}$$

An interior optimal date-specific common price then satisfies

$$\varpi_t = \frac{\mathbb{E}_0[|s_{qq,t}| \varpi_t^{**}]}{\mathbb{E}_0[|s_{qq,t}|]}.$$

All terms now belong to the allocation satisfying the labor, savings, and common-price restrictions jointly (Corollary C.1, Appendix C.3). In particular, the income-wedge multipliers need not equal those obtained when the carbon wedge is free.

Weak separability no longer guarantees that the common rate equals the social cost of carbon. During working life, it leaves $\varpi_t^{**} = -\zeta_t \mathcal{I}_{c,t}^q$. The consumption mix still affects the income-wedge constraints through the marginal utility of the clean good. Retirement removes screening and the labor-wedge constraint, but the savings correction remains. The optimal retirement rate equals the social cost of carbon when this correction has a zero substitution-weighted mean. Disappearing savings multipliers suffice, but cancellation across households can also deliver equality.

A rate held constant over the life cycle must balance marginal resource costs across households as well as dates. When the social cost of carbon is constant across dates, as in the quantitative exercise, let $\varpi = (\bar{\tau}^q - \text{SCC})/(1 + \bar{\tau}^q)$. An interior optimal rate satisfies

$$\varpi = \frac{\sum_{t=0}^T R^{-t} \mathbb{E}_0[|s_{qq,t}| \varpi_t^{**}]}{\sum_{t=0}^T R^{-t} \mathbb{E}_0[|s_{qq,t}|]} \quad (12)$$

With income wedges free, replace ϖ_t^{**} by ϖ_t^* . The weights combine compensated demand responses with resource discounting at R^{-t} . A single rate need not satisfy the date-specific optimality conditions: gains at some dates can offset losses at others, including during retirement. If the social cost of carbon varies over time, the optimal rate instead averages $\text{SCC}_t + \bar{p} \varpi_t^{**}$ with the same discounted substitution weights (Corollary C.2, Appendix C.3). Even with weak separability and free income wedges, a constant rate therefore cannot track a changing social cost of carbon.

6 Quantitative analysis

Whether the only thing constraining redistribution is the government’s inability to directly condition taxes on skill or also to optimize the labor and savings margins, the theory shows that the optimal carbon tax is household-specific and age-dependent. However, how much is indeed gained from considering more complex carbon taxation schemes? To answer this question, I compare the two problems calibrated to US data. First, I consider the case of a planner choosing an allocation that optimizes the labor, savings, and carbon wedges jointly. Second, I consider the case of a planner that can only move the carbon wedge, the labor wedge and the savings wedge fixed at the schedules implied by the US tax system.

6.1 Model and calibration

The lifecycle model and most of the computational method follow Farhi and Werning (2013), expanded to two goods. Agents enter the workforce at age 25, work to age 64, and then retire from age 65 to 84. Both retirement and death are fixed. Individual log skills follow a random walk. To simplify computation, households are identical when they start working. Each period, households consume a clean numéraire and a composite energy good, both with price one. Utility is

$$U(c, q, l) = \frac{1}{\rho} \log[(1 - b_q)c^\rho + b_q e^{-\varkappa l^\alpha} (q - \bar{q})^\rho] - \frac{l^\alpha}{\alpha} \quad \rho \equiv 1 - \frac{1}{\varsigma}$$

The homothetic benchmark sets $\bar{q} = 0$. Positive $\varkappa$ makes energy a relative leisure complement, the case in which optimal income taxation causes a positive carbon-price premium. The energy share is 7%, the substitution elasticity is $\varsigma = 0.4$, and $\varkappa = 0.25$ targets a mid-career energy wedge of about 10% when the externality is absent. A Stone-Geary variant sets $\bar{q} > 0$ to match an income elasticity of 0.45. I use this variant only to compare linear carbon prices. Table 1 summarizes the calibration.

The externality enters as a fixed consumption-unit resource cost $e = 0.494$ per unit of energy. The social resource price is therefore $1 + e$ in every regime whatever the consumer price and the corrective benchmark is $\tau^q = e$. This imports a social cost of carbon of 2020\$190/tCO₂ (U.S. Environmental Protection Agency 2023), using a fixed carbon intensity of 0.0026t/\$. The social cost of carbon is fixed, omitting the endogenous pollution-stock dynamics of Section 4.3. I hold marginal climate damages fixed to isolate the role of redistribution in shaping carbon taxes. The exercise therefore studies tax design within a cohort, taking the social cost of carbon as given.

The fixed labor schedule uses progressivity parameter $p = 0.181$ from Heathcote et al. (2017), evaluated in the benchmark’s normalized earnings. A capital income tax of $\tau^k = 0.18$ implies a savings wedge of $\bar{\tau}^s = 0.009$ by (44). The capital tax is based on the Congressional Budget Office

(2014) report. Appendix D.1 explains the mapping from observed taxes to these restrictions.

Table 1: Calibration

Parameter	Value	Moment or source
<i>Life-cycle block (Farhi and Werning 2013)</i>		
Discount factor $\beta = 1/R$	0.95	$\beta R = 1$
Labor-disutility curvature α	3	Baseline Frisch elasticity 0.5
Log-skill innovation variance $\hat{\sigma}^2$	0.0095	US male wages
Working / retirement periods	40/20	Ages 25–64 / 65–84
<i>Two-good block</i>		
Energy expenditure share	0.07	Consumer Expenditure Survey, 2022–23
Substitution elasticity ς	0.4	Implied own-price elasticity about 0.44
Energy income elasticity	1 / 0.45	Homothetic / Stone-Geary calibration
Leisure complementarity $\varkappa$	0.25	Mid-career zero-externality wedge about 10%
Marginal damage e	0.494	$\$190/\text{tCO}_2 \times 0.0026 \text{ tCO}_2/\$$
<i>Fixed schedules, Appendix D.1</i>		
Labor progressivity p	0.181	Heathcote et al. (2017)
Capital income tax τ^k	0.18	Effective marginal rate, 2014 law (Congressional Budget Office 2014)
Implied savings wedge $\bar{\tau}^s$	0.9% per year	(44) at $\beta R = 1$

6.2 Carbon wedge over the life cycle

With optimal income wedges, the average carbon price rises during working life and returns to the Pigouvian level at retirement (Figure 1). With income wedges fixed at the calibrated US schedules, it instead declines with age and becomes negative late in working life. In the first regime, dispersion widens as skill shocks accumulate during working life, then disappears at retirement because there is no further screening motive.

Average carbon rates rise with current skill in both regimes, but much more steeply with fixed income wedges (Figure 2). At age 64, the gap between the lowest and highest current-skill quintile means is about 14 percentage points with optimal income wedges and 44 percentage points with fixed wedges, more than three times as large.

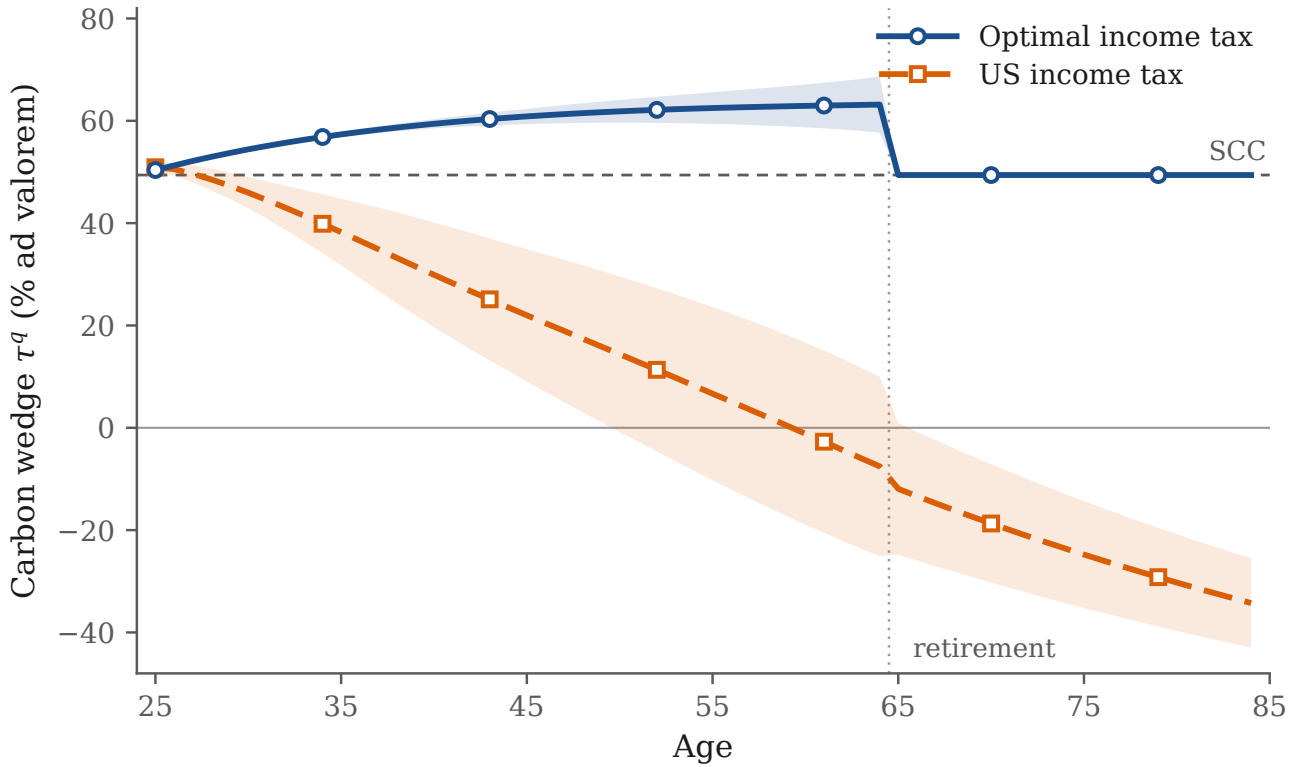


Figure 1: Average carbon wedges with optimal and US-calibrated fixed wedges. Bands show ± 1 cross-sectional standard deviation. The horizontal line is the Pigouvian rate, $e = 0.494$.

Figure 3 compares the allocations obtained by imposing the two income-wedge constraints sequentially. I first fix the labor schedule while leaving savings free, then add the savings-wedge constraint, re-optimizing after each restriction.

Fixing the labor schedule lowers average carbon prices early in working life and raises them later. Adding the savings constraint reverses the upward age profile and produces negative carbon wedges late in life. Although the annual savings wedge is only $\bar{\tau}^s = 0.009$, its effect compounds, requiring mean marginal utility of clean consumption to rise by about 70% between ages 25 and 84.

Retirement eliminates the screening motive but leaves the savings constraint in place. The imposed capital tax continues to distort consumption over time, although the optimal retirement savings wedge is zero. Carbon pricing partly offsets this distortion; in the benchmark, the correction lowers average carbon prices further during retirement.

The average carbon price declines because the savings constraint dominates in this calibration. The theory does not prescribe this age pattern.

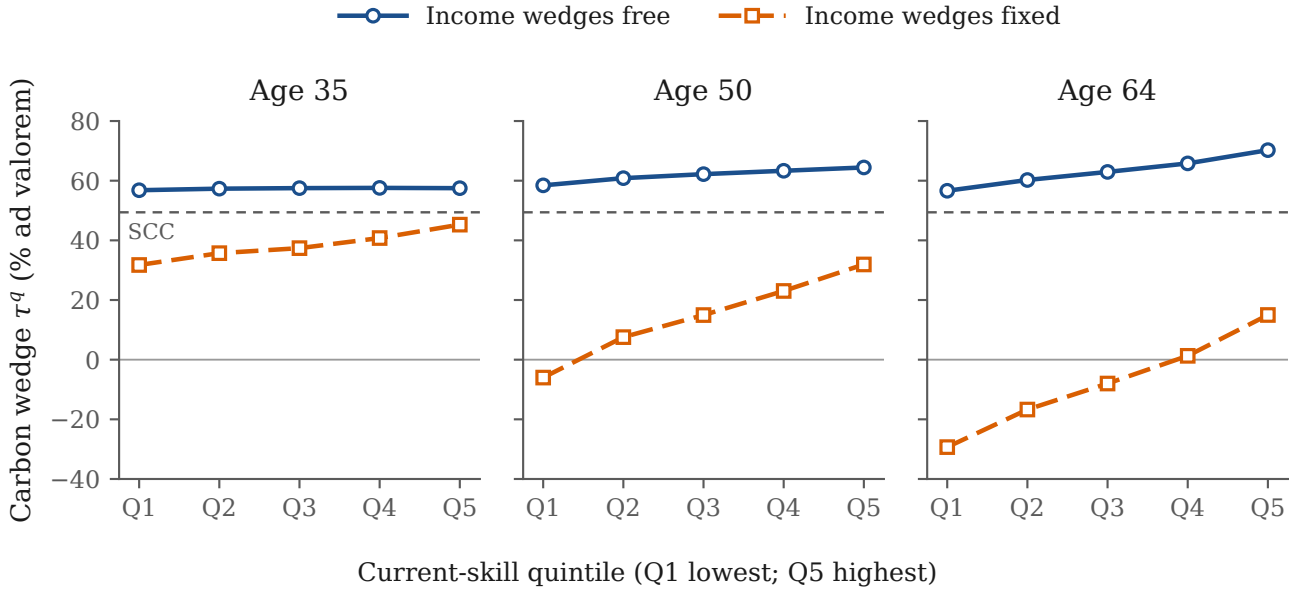


Figure 2: Average carbon wedges by current-skill quintile, with income wedges free or fixed. Each panel compares the two allocations at one age on the same vertical scale. Dashed horizontal lines mark the Pigouvian rate.

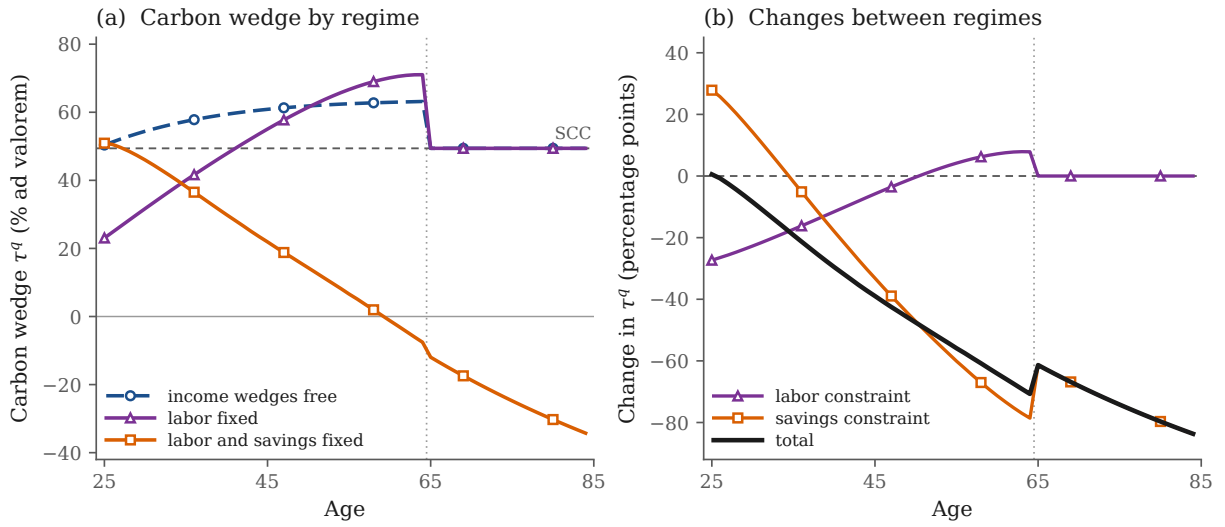


Figure 3: Carbon prices when the labor wedge is fixed first, followed by the savings wedge implied by $\tau^k = 0.18$. Panel (a) gives the three policy profiles; panel (b) gives their consecutive differences, whose split depends on this order. Retirement carbon prices with savings free are analytical.

6.3 The value of carbon-price differentiation

Carbon-price differentiation adds little welfare when income wedges are free and much more when they are fixed. Table 2 compares linear, age-dependent, and history-dependent carbon prices. Each column measures gains over its income-tax regime's own no-carbon allocation. The comparisons concern the relaxed allocation problems of Section 5.1.2, with transfers re-optimized but they do not by themselves establish gains from implemented tax reforms.

6.3.1 Linear versus nonlinear carbon pricing

With income wedges free, income taxation supplies insurance and redistribution, leaving the common carbon price to charge for the externality. Differentiation can improve screening through energy consumption, but this additional role is small in the benchmark.

The full optimum gains 0.313% of consumption over no carbon price. A linear tax at the social cost of carbon gains 0.309%, about 99% of that value, and cuts emissions by 13.5%. Optimizing the linear rate raises it from $e = 0.494$ to 0.555, or from \$190 to \$214 per ton, but adds only 0.002 percentage points of consumption. Using the full optimum's mean wedge at each age gives 0.312%. Each successive step toward a more flexible price rule adds at most 0.002 points, below the comparison's 0.003-point numerical resolution. The corrective rate captures nearly all the measured gain, while the small differences among the optimized rules remain unresolved.

Subsistence energy demand reduces the abatement from linear pricing. The optimized linear tax cuts emissions by 6.3%, and its best rate remains close to 0.56. Its welfare gain is 0.152% in units of consumption above subsistence. This measure scales the clean good and energy consumption net of $\bar{q}$, rather than total consumption. Table 6 in Appendix D.4 reports this preference comparison and the checks without an externality.

With income wedges fixed at the benchmark schedules, the best linear rate falls to 0.121, about \$47 per ton, and gains only 0.021% of consumption. Charging the social cost of carbon instead loses 0.115%, even after accounting for avoided damages. These results retain the benchmark's earnings-normalization qualification in Section 5.1.2.

Allowing history-dependent carbon wedges raises the gain to 1.401% against the same no-carbon baseline, an improvement of 1.38 percentage points over the best linear rate. Differentiation lets the carbon wedge partly offset labor and savings distortions, with corrections that vary across household histories and ages. A common rate must balance these corrections, including those in retirement (Corollary C.2). At the corrective rate, the costs at the constrained margins outweigh the environmental gains. The history-dependent allocation also cuts discounted emissions more than the best linear rate, by 8.0% against 4.3%.

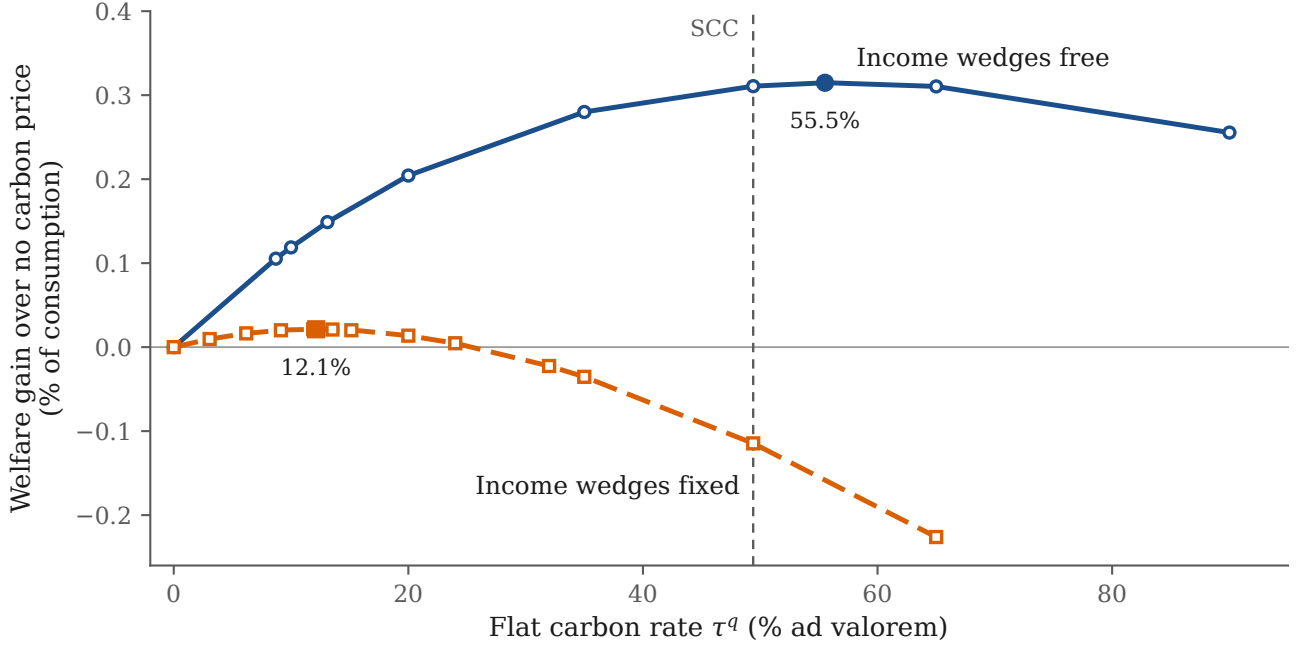


Figure 4: Welfare gains from a linear carbon tax with income wedges free or fixed, relative to each regime’s own no-carbon allocation. The vertical distance compares reform gains, not welfare levels across regimes. Both curves use the same coarse grid and simulated histories; Table 2 uses a finer grid for its free-income column. Filled markers identify the best rates, and the vertical dashed line marks the social cost of carbon.

Under the alternative labor calibration (Section 6.5), the best linear rate rises to \$187 per ton, while the nonlinear gain remains large at 1.744% (Table 5). Price flexibility still adds substantial value, but the linear corrective rate now raises welfare.

Table 2: Welfare value of carbon pricing: income wedges free or fixed

Carbon-price rule	Income wedges free	Income wedges fixed
linear rate at the social cost of carbon	0.309	−0.115
Optimal linear rate	0.311	0.021
Age-varying common rate	0.312	0.145
History-dependent wedge	0.313	1.401

Note: consumption-equivalent gains in % of consumption over each column’s no-carbon-price allocation. The free-income column uses the finer-grid comparison of Table 6. Differences below 0.003 percentage points are unresolved. Its age-varying rate is the full optimum’s mean wedge at each age. The age-varying entry with fixed income wedges uses the scaling factor $a = 1.25$ defined in Section 6.3.2. Appendix D.4 gives the details. Transfers are re-optimized in each allocation problem.

6.3.2 Age-dependent carbon taxes

Age-dependent linear taxes recover part of the gain of fully nonlinear taxation while charging every household of a given age the same rate. They accommodate the life-cycle pattern of the income-wedge corrections without conditioning the rate on individual histories. Age and calendar time coincide in this single-cohort model. Applying the policy to overlapping cohorts would require distinguishing purchases by age.

I construct a candidate age profile r_t from the history-dependent allocation's mean wedge at each working age and a single mean rate in retirement. I then consider schedules $\tau_t^q = e + a(r_t - e)$, which multiply each age's deviation from the social cost of carbon by a common factor a . Among the values considered, $a = 1.25$ gives the highest computed welfare. This schedule, shown in Figure 5(a), gains 0.145% of consumption and cuts emissions by 6.36%. It yields almost seven times the gain of the best linear rate and closes about 9% of its gap with the history-dependent allocation.

Simpler designs also improve on linear pricing. Among the schedules considered with one working-life rate and one retirement rate, the largest computed gain is 0.063%, while a straight declining working-age profile gains 0.126%. Gains remain when subsidies are ruled out: flooring a matched candidate at zero gives 0.109%, against 0.129% for its unfloored counterpart, and raises abatement. The estimated welfare cost of the floor, about 0.02 percentage points, is only modestly larger than the observed variation across solution chains.

Figure 5(b) instead scales the entire candidate profile. It compares age-dependent and linear prices using the same fixed benchmark emissions weights to summarize their mean rates. Within this family, age differentiation preserves positive gains at mean rates near the social cost of carbon, where linear pricing lowers welfare. Equal weighted mean rates do not imply equal emissions or revenue. Appendix D.5 defines the searches and reports the simpler candidates.

The searches establish gains from age differentiation but do not identify the optimal age schedule. A two-parameter search improves the reported gain to 0.154%, at a boundary point. Its advantage over scaling deviations from the social cost of carbon alone is below the observed numerical variation across solution chains. The remaining gap to the history-dependent allocation therefore combines possible gains from better age schedules with gains from conditioning on household histories. As with the other comparisons, the accompanying transfers remain subject to implementation checks.

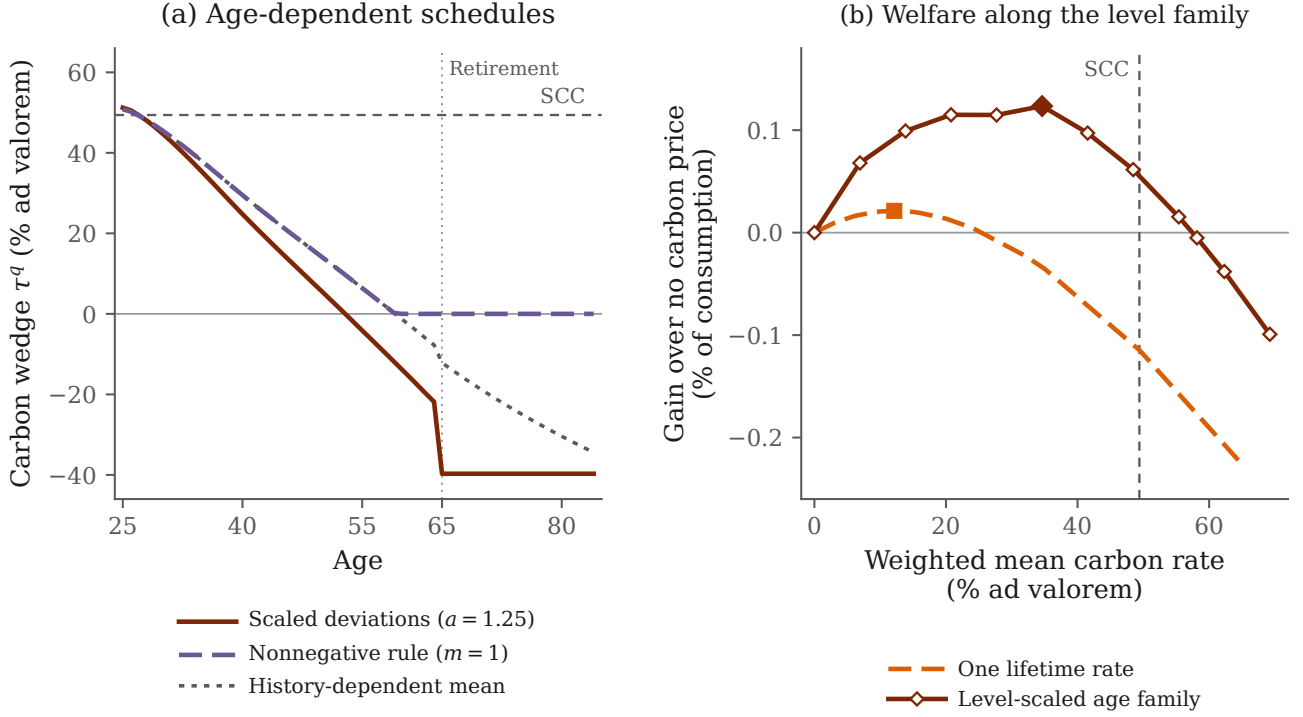


Figure 5: Age-dependent linear carbon taxes with both income wedges fixed. Panel (a) compares the schedule with deviations from the social cost of carbon scaled by $a = 1.25$, the nonnegative candidate, and the history-dependent allocation's mean wedge. Panel (b) scales the entire candidate profile by m and compares it with linear pricing, using fixed benchmark discounted-emissions weights to summarize each schedule. Welfare is relative to the common no-carbon allocation.

6.4 Transfers and labor-tax cuts

At the best linear carbon rate, the implied transfer increment is nearly uniform across households. Allowing the labor schedule to adjust instead raises welfare substantially in the benchmark. These two comparisons show how the available income-tax instruments affect both the aggregate gain and its distribution.

6.4.1 Implied rebates

The implied lifetime transfer $B(\theta^T)$ is the present value of spending at consumer prices less retained earnings under the normalized income schedule. Both use the household's after-tax return for discounting (Appendix D.6). This accounting residual is the transfer needed to support the allocation's household budget. The baseline transfer is strongly redistributive, averaging 19.4% of lifetime spending in the bottom earnings quintile and 2.1% in the top.

At the best linear rate, the mean transfer increment is 97.5% of mean discounted carbon payments. This ratio uses the household’s discount rate, rather than the resource discount rate used in government accounts. The increment’s standard deviation is only 2.4% of its mean and the remaining variation has not been distinguished from numerical error. It resembles a uniform lifetime dividend even though baseline transfers vary substantially across households.

The dividend allocation favors lower earners in the benchmark (Table 3). Carbon payments rise with lifetime spending, while the transfer increment changes little. At the best linear rate, the bottom quintile gains 0.27% of consumption on average and the top loses 0.19%. Even at the corrective rate, which lowers aggregate welfare, the bottom two quintiles gain on average. These findings concern the homothetic benchmark where lifetime energy shares are nearly constant across quintiles. Incidence under subsistence demand has not been computed.

Table 3: Implied transfers and welfare changes by lifetime-earnings quintile

quintile	B_0	best flat rate, \$47/t			Pigouvian, \$190/t		
		rebate	paid	ΔCE	rebate	paid	ΔCE
1 (lowest)	19.4	0.97	0.74	+0.27	0.96	0.74	+0.77
2	13.6	0.97	0.86	+0.10	0.96	0.87	+0.17
3	10.1	0.97	0.96	+0.01	0.96	0.96	−0.16
4	6.8	0.98	1.08	−0.08	0.97	1.08	−0.47
5 (highest)	2.1	0.98	1.35	−0.19	0.97	1.34	−0.87

Note: benchmark fixed schedules, $\tau^k = 0.18$. Quintiles use pre-reform lifetime earnings. B_0 is the no-carbon allocation’s implied transfer in % of lifetime spending, both discounted at the after-tax return. *Rebate* is the transfer increment and *paid* the carbon payments, each divided by mean carbon revenue per history. ΔCE is the quintile’s mean welfare change in % of consumption, from paired histories.

6.4.2 Recycling into rate cuts

The tax swap changes the level of the labor schedule while keeping the savings wedge and the labor progressivity parameter fixed. It scales the net-of-tax labor schedule by a common factor.

$$1 - \bar{\tau}_t^y \mapsto s(1 - \bar{\tau}_t^y),$$

It chooses s so that the mean implied lifetime transfer remains at its no-carbon level. This is the mean-transfer neutrality condition used in the exercise. Transfers are still re-optimized and can vary across histories.

At the social cost of carbon, the swap gains 0.432% of consumption, compared with the dividend allocation’s loss of 0.115% (Figure 6). The labor-rate cut is 2.8 percentage points at the mean

marginal rate. The largest estimated gain on the reported grid is 0.465%, at a carbon rate of 0.65 (Table 8). Allowing labor-rate cuts therefore raises the gains from carbon pricing and makes higher linear rates desirable in this benchmark. This gain includes the benefit of changing the income-tax schedule.

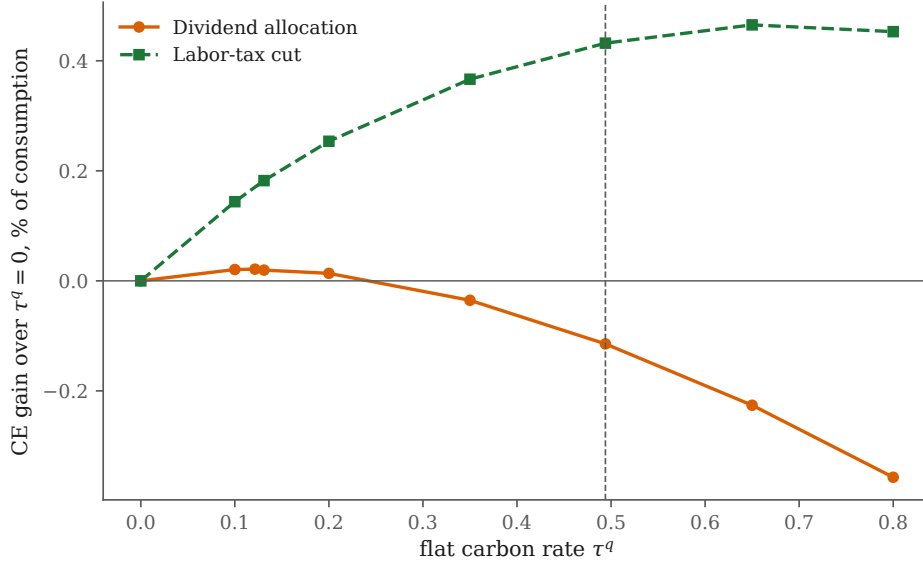


Figure 6: Welfare gains from the dividend and labor-rate-cut allocations, relative to the same no-carbon baseline. Savings wedges remain fixed in both; the swap relaxes the labor-wedge constraint while holding the mean implied transfer fixed. The vertical dashed line marks the social cost of carbon. Curves interpolate the comparisons in Table 8.

The rate cut relaxes the large labor distortions at young ages, although cutting the same schedule later in life can be costly. Its net value depends on how the prescribed schedule differs from the desired one. Under the alternative labor calibration, the swap’s advantage over the dividend at the social cost of carbon falls from 0.55 to about 0.04 percentage points.

The swap also changes who gains. At the social cost of carbon, average consumption-equivalent gains are positive in every baseline lifetime-earnings decile, from 0.28% at the bottom to 0.55% at the top. Positive decile averages do not rule out individual losses however. At this rate, the dividend favors poorer households more, while rate cuts benefit richer households more. Recycling choices affect both welfare and distribution, as in Goulder et al. (2019), Klenert et al. (2018), and Fried et al. (2024). The rate cut also expands labor and spending, reducing abatement from 14.3% under the dividend to 13.1% under the swap. Appendix D.6 reports the neutrality checks, rate grid, and incidence sensitivity.

6.5 Scope and sensitivity

The comparison establishes a conditional role for carbon-price differentiation. With income wedges free, a linear corrective tax captures almost all the measured gain in the benchmark. With income wedges fixed, the commodity margin can also repair labor and savings distortions, making flexibility much more valuable. The level of the best linear rate depends on the fixed schedules and on what the revenue may finance. The benchmark's low rate is therefore not a general prescription for an economy with distortionary income taxes.

The labor calibration is especially consequential. Households are assumed to be identical when entering the workforce, so there is not a lot of redistribution to be done early and the labor wedge is near zero. This makes the fixed labor schedule relatively distortionary at young ages. An alternative calibration instead shifts the fixed schedule to reproduce the labor gaps implied by the wedge endpoints of Golosov et al. (2016), about 25% early in life and 65% late. It retains this paper's skill process and preferences and does not reproduce their economy with initial heterogeneity. The labor-wedge constraint then raises the summary carbon price, whereas adding the savings-wedge constraint lowers it under both calibrations (Table 5). Hence, the sign of the labor effect is sensitive to the desired labor schedule. The savings-multiplier sign reversal in Lemma 5.2 remains under its stated assumptions, but neither the carbon price's crossing date nor its direction of change is fixed by that lemma.

The damage calibration also matters for more than dollar units. Re-solving the homothetic model with $e = 0.133$ reduces the full-optimum gain from 0.313% to 0.040% of consumption. A linear tax at the social cost of carbon then captures about 89% of the gain, compared with 99% at the benchmark. The best linear rate captures about 95%. These shares are approximate as the associated welfare gaps, 0.0043 and 0.0022 percentage points, are close to the numerical discrepancies observed in the benchmark. The point estimates suggest that differentiation becomes more valuable relative to corrective benefits as damages fall, while its absolute value stays small. Dollar conversion is proportional at a fixed calibration. Changing marginal damage changes the allocation and requires resolving the model.

Three restrictions limit extrapolation. First, relative leisure complementarity determines the sign of the premium when income taxes are optimal. Its magnitude is calibrated from gasoline evidence and is less well pinned down for the broader energy composite. Second, the unrestricted commodity solver requires a homothetic log consumption index, so the necessity-demand comparison does not establish the value of nonlinear pricing under nonhomothetic preferences. Third, the fixed savings wedge continues through retirement. This restriction contributes to the late-life carbon subsidy and to the low best linear rate. The numerical exercise illustrates these mechanisms within a single-cohort, single-intensity energy model. It does not estimate an implementable US reform or the welfare gain from changing the income-tax system itself.

7 Conclusion

I modify the Golosov et al. (2016) dynamic Mirrlees model to feature two goods, one clean and one dirty, to study optimal carbon taxation with income inequality and nonlinear taxation. Extending the Jacobs and Mooij (2015) irrelevance result to a dynamic setting, I find that correction and redistribution are separate when labor is separate with consumption and pollution damages in utility. More specifically, the dirty good is overdistorted when it is a labor substitute and underdistorted when it is a labor complement, with the optimal commodity wedge being not only skill-dependent but also age-dependent.

In the quantitative benchmark, a linear tax at the social cost of carbon captures about 99% of the measured welfare gain when income wedges are chosen optimally. In other words, a nonlinear carbon tax has little value over an optimally-set linear tax when the rest of the tax system is already optimal. However, holding labor and savings wedges fixed may make instrument flexibility more valuable. Calibrating to US data, the best linear rate falls to about \$47tCO₂ and captures only 1.5% of the gain of a nonlinear tax. Age-dependent linear taxes offer an intermediate instrument, which is easier to implement since dirty good purchases need not be tracked and since age-dependent policies are plausible⁵. Scaling the candidate age profile's deviations from the social cost of carbon by 1.25 gains 0.145% of consumption against 0.021% for the best linear rate, while a nonnegative candidate still gains 0.109%.

The level of the preferred linear rate depends on the labor calibration and on revenue use. An alternative calibration brings it close to the social cost of carbon. Allowing revenue to finance marginal labor-rate cuts also changes the benchmark comparison. At the social cost of carbon, a tax swap gains 0.432% of consumption, while the allocation with fixed income wedges and an approximately uniform transfer increment loses 0.115%. The linear rate and the income-tax margins open to reform must therefore be evaluated together. The benchmark's low rate is not a general implication of distortionary income taxation.

A number of questions remain open to future research. First, this work focuses on optimal allocations rather than the ways to implement them in an actual tax system. Moreover, the wedge restriction is a strong assumption, making more flexible ways to bypass income tax constrains a natural next step. For example, performing more thorough analysis on carbon revenue taxation and the best ways it can be recycled. Second, overlapping generations would allow pollution dynamics to link emitting cohorts to those bearing future damages and distinguish age-dependent taxes from a common price that changes over time. Moreover, this would allow to compute the social cost of carbon rather than rely on an external benchmark.

⁵Consider that youth, student, and senior discounts are already common practice for example.

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A Optimal wedge derivations

A.1 The problem and first-order conditions

Write $l = y/\theta$, so $U_\theta = -U_l l/\theta$, and let $f_2 = \partial f_t / \partial \theta_{t-1}$. Suppressing dates and histories where unambiguous, the Hamiltonian of Section 4.1 is

$$\begin{aligned} \mathcal{H} = & [c + q - \theta l + R^{-1}V_{t+1}(w, w_2, \theta, \mathbf{S})] f + \psi \left[-U_l \frac{l}{\theta} + \beta w_2 \right] \\ & - \lambda_1 u \bar{\alpha}_t f + \lambda_2 u f_2 + \phi[u - U - \beta w] - \mu[S - S(\dots, Q)]. \end{aligned}$$

At an interior allocation its conditions are

$$\begin{aligned} f - \psi U_{cl} \frac{l}{\theta} &= \phi U_c, & (1 + \nu_t) f - \psi U_{ql} \frac{l}{\theta} &= \phi U_q, \\ -\theta f - \phi U_l &= \frac{\psi}{\theta} (U_{ll} l + U_l), & \dot{\psi} &= \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi, \\ R^{-1} V_{w,t+1} f &= \beta \phi, & R^{-1} V_{w_2,t+1} f &= -\beta \psi, \\ \mu_t &= \int \left(-\psi U_{Sl} \frac{l}{\theta} - \phi U_S \right) d\theta. \end{aligned}$$

The costate satisfies $\psi(0) = \psi(\infty) = 0$. The shadow price of current emissions is

$$\nu_t \equiv \mu_t S_q + \frac{1}{R} \frac{\partial V_{t+1}}{\partial Q_t}, \quad S_q \equiv \frac{\partial S_t}{\partial Q_t}. \quad (13)$$

The extra density in the q condition comes from $Q_t = \int q f d\theta$. Current emissions enter continuation costs through the updated emission window $\mathbf{S}$; the scalar current stock S is not a separate continuation argument. Aggregate equations integrate over the population of histories, including the distribution of prior histories suppressed in the component notation.

A.2 Validity of the recursive formulation

Proposition A.1 (Recursive formulation with an externality).

Suppose the two-good economy satisfies the regularity conditions for the local recursive reduction of Kapička (2013). With deterministic aggregates, pollution adds the emission window $\mathbf{S}_- = (Q_{t-s}, \dots, Q_{t-1})$ and aggregate consistency constraints. Given the emission price ν_t , the component allocation problem has dated utility $U(c, q, l, S_t)$ and dirty-good resource cost $1 + \nu_t$. The pollution multiplier μ_t and emission price are common across date- t histories, with

$$\nu_t = \sum_{j=0}^s R^{-j} \mu_{t+j} \Phi_j, \quad \Phi_j \equiv \frac{\partial S_{t+j}}{\partial Q_t}.$$

Multipliers beyond the terminal date are zero. The recursive and sequence formulations give the same first-order conditions.

Proof. An individual report has measure zero in aggregate emissions, so every reporting strategy faces the same pollution path. Defining $\tilde{U}_t(c, q, l) = U(c, q, l, S_t)$ therefore leaves every global incentive comparison unchanged. Under the differentiability and envelope conditions of the maintained reduction, the necessary local constraint is

$$\dot{u}_t = U_\theta + \beta w_{2,t} = -U_l \frac{l_t}{\theta_t} + \beta w_{2,t}.$$

There is no U_S term because an individual's type does not change S_t . Fixing the stock does not itself supply missing conditions of the reduction: its compactness, integrability, and relevant concavity requirements remain maintained assumptions. In particular, where joint concavity in (c, q, l) is required, it is additional to Assumption 3.1.

To internalize pollution, write $Q_t = \mathbb{E}_0[q_t(\theta^t)]$ and $S_t = S(Q_{t-s}, \dots, Q_t)$ as separate constraints. With $\mathcal{L}^{\text{alloc}}$ the local allocation Lagrangian at a fixed stock path, the sequence Lagrangian is

$$\begin{aligned} \mathcal{L} = \mathcal{L}^{\text{alloc}} &+ \sum_{t=0}^T R^{-t} \nu_t (\mathbb{E}_0[q_t] - Q_t) \\ &- \sum_{t=0}^T R^{-t} \mu_t [S_t - S(Q_{t-s}, \dots, Q_t)]. \end{aligned}$$

For fixed aggregates and prices, its allocation terms have flow resource cost $c_t + (1 + \nu_t)q_t - y_t$. Thus the original component recursion applies with dated utility and a linear price adjustment. Where the reduction requires an increasing resource cost, maintain $1 + \nu_t > 0$. No stationarity is needed over the finite horizon.

Stationarity in S_t gives the population damage integral in Appendix A.1. Stationarity in Q_t gives

$$-R^{-t} \nu_t + \sum_{j=0}^s R^{-(t+j)} \mu_{t+j} \Phi_j = 0,$$

which establishes the emission-price formula. The maintained absence of aggregate risk makes these prices deterministic population objects, not forecasts conditional on one household's history.

Past emissions affect future utility only through $\mathbf{S}_-$. Appending Q_t and dropping Q_{t-s} therefore supplies the pollution state transition. Component recursions coupled by aggregate consistency reproduce the sequence conditions: the q condition adds $\nu_t f$, the S condition aggregates damages, and the remaining conditions have their original form. This establishes the recursive representation of the local problem. $\square$

A.3 Labor wedge

The c and l conditions give two expressions for the utility multiplier,

$$\phi = \frac{f}{U_c} - \frac{\psi}{\theta} \gamma_c = -\frac{\theta f}{U_l} - \frac{\psi}{\theta} (1 + \epsilon). \quad (14)$$

Using $1 - \tau^y = -U_l/(\theta U_c)$ and equating them yields

$$\frac{\tau_t^y}{1 - \tau_t^y} = A_t \frac{\psi U_{c,t}}{\theta f_t}, \quad A_t \equiv 1 + \epsilon_t - \gamma_{c,t}. \quad (15)$$

Substitution into the costate equation gives

$$\dot{\psi} - \frac{\gamma_c}{\theta} \psi = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \frac{f}{U_c}.$$

Define its integrating kernel $K_t(\theta, x) = \exp[-\int_\theta^x \gamma_{c,t}(z) dz/z]$. Integrating from θ to ∞ and using the upper boundary condition gives

$$\psi_t(\theta) = \int_\theta^\infty K_t(\theta, x) \left[\frac{f_t(x)}{U_{c,t}(x)} - \lambda_{1,t} \bar{\alpha}_t(x) f_t(x) + \lambda_{2,t} f_{2,t}(x) \right] dx. \quad (16)$$

Under the maintained boundary and integrability conditions, $\psi(0) = 0$ pins down

$$\lambda_{1,t} = \frac{\int_0^\infty K_t(0, x) [f_t(x)/U_{c,t}(x) + \lambda_{2,t} f_{2,t}(x)] dx}{\int_0^\infty K_t(0, x) \bar{\alpha}_t(x) f_t(x) dx}.$$

To recover the $ABCD$ formula, split (16) into its redistributive and persistence terms. Since the pollution stock is common across types and $l = y/\theta$,

$$\frac{d}{d\theta} \ln U_{c,t} = -\sigma_t \frac{\dot{c}_t}{c_t} - \sigma_{cq,t} \frac{\dot{q}_t}{q_t} + \gamma_{c,t} \left(\frac{\dot{y}_t}{y_t} - \frac{1}{\theta} \right).$$

Consequently,

$$\frac{U_{c,t}(\theta)}{U_{c,t}(x)} K_t(\theta, x) = \exp \left[\int_\theta^x \left(\sigma_t \frac{\dot{c}_t}{c_t} + \sigma_{cq,t} \frac{\dot{q}_t}{q_t} - \gamma_{c,t} \frac{\dot{y}_t}{y_t} \right) dz \right],$$

where the integrand is evaluated at z . Substituting this identity into the first term of (16) gives $A_t B_t C_t$, with B_t and C_t defined in Proposition 4.1.

For the persistence term, the w_2 condition and the envelope $V_{\hat{w}_2,t} = -\lambda_{2,t}$ imply

$$\lambda_{2,t} = \beta R \frac{\psi_{t-1}}{f_{t-1}} = \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} \frac{\theta_{t-1}}{A_{t-1} U_{c,t-1}}.$$

Its contribution to (15) is therefore

$$\frac{A_t U_{c,t}}{\theta f_t} \lambda_{2,t} \int_{\theta}^{\infty} K_t(\theta, x) f_{2,t}(x) dx = \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} D_t(\theta).$$

Adding the two terms proves

$$\frac{\tau_t^y}{1 - \tau_t^y} = A_t B_t C_t + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} D_t.$$

At the initial date the inherited persistence term is absent. At the retirement junction the continuation restriction is the one in Appendix A.7.

A.4 Savings wedge

Let $\chi_t = \gamma_{c,t} \psi U_{c,t} / (\theta f_t)$, equivalently the incentive credit in Proposition 4.1. The c condition becomes

$$\frac{\phi}{f} = \frac{1 - \chi_t}{U_{c,t}}.$$

The w condition and the envelope $V_{\hat{w},t} = \lambda_{1,t}$ then give

$$\lambda_{1,t+1} = \beta R \frac{1 - \chi_t}{U_{c,t}}. \quad (17)$$

Integrating the costate equation, using its two boundary conditions, $\int \bar{\alpha}_t f_t d\theta = 1$, and $\int f_{2,t} d\theta = 0$, also gives

$$\lambda_{1,t} = \int \phi d\theta = \mathbb{E}_{t-1} \left[\frac{1 - \chi_t}{U_{c,t}} \right].$$

Apply this expression at $t + 1$ and equate it with (17). At every nonterminal date,

$$\frac{1 - \chi_t}{U_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1 - \chi_{t+1}}{U_{c,t+1}} \right]. \quad (18)$$

Where $1 - \chi_t > 0$, writing $\hat{U}_{c,t} = U_{c,t} / (1 - \chi_t)$ gives the adjusted inverse Euler equation of Proposition 4.1.

A.5 Commodity wedge

Divide the c and q conditions by their marginal utilities, use $U_q / U_c = 1 + \tau^q$, and equate the utility multipliers:

$$1 - \gamma_c \frac{\psi U_c}{\theta f} = \frac{1 + \nu_t}{1 + \tau^q} - \gamma_q \frac{\psi U_c}{\theta f}.$$

Rearranging and applying (15) gives

$$\frac{\tau_t^q - \nu_t}{1 + \tau_t^q} = (\gamma_{c,t} - \gamma_{q,t}) \frac{\psi U_{c,t}}{\theta f_t} = \frac{\gamma_{c,t} - \gamma_{q,t}}{1 + \epsilon_t - \gamma_{c,t}} \frac{\tau_t^y}{1 - \tau_t^y}.$$

Identifying ν_t with the social cost of carbon proves (5). Weak separability sets the complementarity gap to zero and gives Corollary 4.1.

A.6 The social cost of carbon

Substitute (14) into the S condition. Using $\xi_t = U_{Sl,t} l_t / U_{S,t}$ gives the population damage value

$$\mu_t = -\mathbb{E}_0 \left[\left(1 + (\xi_t - \gamma_{c,t}) \frac{\psi U_{c,t}}{\theta_t f_t} \right) \frac{U_{S,t}}{U_{c,t}} \right].$$

The expectation $\mathbb{E}_0$ averages all date- t histories. Equation (15) gives the fiscal-adjustment factor in Proposition 4.2.

The emission-price formula in Proposition A.1 then sums these damage values over the dates affected by current emissions. Equivalently, the continuation envelope gives

$$\frac{\partial V_{t+1}}{\partial Q_t} = \sum_{j=1}^s R^{-(j-1)} \mu_{t+j} \Phi_j, \quad \nu_t = \sum_{j=0}^s R^{-j} \mu_{t+j} \Phi_j,$$

with $\Phi_0 = S_q$ and $\mu_{t+j} = 0$ beyond the terminal date. Substituting the damage value establishes (6). Removing the fiscal adjustment gives (7); equality follows when $\xi = \gamma_c$ at all affected histories or when their labor wedges vanish.

A.7 Retirement

Lemma A.1 (Retirement and the costate).

An optimal retirement allocation can be taken measurable with respect to the retirement-entry history θ^{T_R-1} . The continuation derivative satisfies $w_{2,t} = 0$ for $t \geq T_R - 1$. The reduced retirement problem is deterministic within each entry history and carries no reporting costate, so $\psi_t = 0$ for $t \geq T_R$. At the last working date, $\dot{u}_{T_R-1} = U_\theta$.

Proof. At a retirement date, labor is zero and current skill enters neither preferences nor production. Every true type therefore ranks reports by the same payoff,

$$U(c(\hat{\theta}), q(\hat{\theta}), 0, S_t) + \beta w(\hat{\theta}).$$

Global incentive compatibility forces this payoff to be constant across reports. Every draw faces the same least-cost delivery problem for that utility, so a common assignment can be

chosen. Backward induction makes the retirement allocation depend only on the history at entry, without subsequent skill risk.

Since next period's utility is constant in its new draw, for $t \geq T_R - 1$,

$$w_{2,t} = \bar{u}_{t+1} \int \partial_{\theta_t} f_{t+1}(\theta' \mid \theta_t) d\theta' = 0.$$

At the last working date this is a restriction on feasible continuation promises, not a first-order condition for a freely chosen w_2 . The future-rent term disappears from its envelope condition, while its current costate remains free.

For each retirement-entry history the component problem reduces to

$$V_t(\hat{w}, \mathbf{S}_-) = \min_{c,q,w,S} \{c + q + R^{-1}V_{t+1}(w, \mathbf{S}) : \hat{w} = U(c, q, 0, S) + \beta w\},$$

coupled to the aggregate emission constraints. Reporting and threat keeping are absent, so their multipliers are zero in this reduced representation. The reporting restriction is essential: with persistent skills, a problem imposing only the local envelope condition could use payoff-irrelevant retirement reports to alter promised utility and relax earlier incentive constraints. Global incentive compatibility rules out that construction.

At $l = 0$, the products U_{cl} , U_{ql} , and U_{Sl} vanish. The retirement conditions therefore have no screening term, give the ordinary inverse Euler equation, and price carbon at Pigouvian damages when income wedges remain free. Restrictions on income wedges are treated separately in [Appendix B](#). $\square$

B Derivations with fixed income wedges

B.1 Recursive formulation and first-order conditions

The additional state. The income-wedge restrictions limit allocations without changing a household's payoff from a reporting strategy. Under the first-order approach maintained in Appendix A.2, the one-shot-deviation reduction and envelope condition therefore remain those of Fernandes and Phelan (2000) and Kapička (2013). The additional restrictions do not establish global incentive compatibility.

The labor-wedge constraint involves only the current bundle and pollution stock, so it adds a pointwise multiplier and no state. For savings, define at each nonterminal history

$$M(\theta^t) \equiv \frac{U_{c,t}(\theta^t)}{(1 - \bar{\tau}_t^s(\theta_t))\beta R}.$$

Constraint (9) requires $\mathbb{E}_t[U_{c,t+1}] = M(\theta^t)$. This requirement need not be determined by the existing promises (w, w_2) , so it enters the successor's state as $\hat{M}$. The successor delivers it through the constraint

$$\int U_c(c(\theta), q(\theta), y(\theta)/\theta, S) f_t(\theta \mid \theta_-) d\theta = \hat{M}.$$

Attaching multiplier ϑ_t gives $V_{t,\hat{M}} = -\vartheta_t$. There is no inherited requirement at birth and no successor at the terminal date, so $\vartheta_0 = \vartheta_{T+1} = 0$.

An allocation satisfying the Euler equation with the prescribed savings wedge supplies each successor with a requirement it meets. Conversely, component allocations meeting these requirements satisfy that equation at every history. This establishes the recursive encoding in Lemma 5.1. Neither restriction directly contains u, w , or w_2 , so their first-order and envelope conditions, including the boundary conditions for ψ , retain their form. The restrictions change ϕ and hence the solution of those conditions.

Allocation conditions. The Hamiltonian augments that of Appendix A.1 with the labor constraint and the marginal-utility requirement:

$$\begin{aligned} \mathcal{H} = & \left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, M(c, q, l, S), \theta, \mathbf{S}) \right] f \\ & + \psi \left[-U_l \frac{l}{\theta} + \beta w_2 \right] - \lambda_1 u \bar{\alpha}_t f + \lambda_2 u f_2 + \phi[u - U - \beta w] - \mu[S - S(\dots, Q)] \\ & + \eta \left[-\frac{U_l}{\theta U_c} - (1 - \bar{\tau}_t^y(\theta)) \right] f + \vartheta_t (U_c - \hat{M}) f. \end{aligned}$$

For a control $z \in \{c, q, l, S\}$, the inherited requirement and continuation value together contribute

$$\left[\vartheta_t - \frac{\vartheta_{t+1}}{(1 - \bar{\tau}_t^s)\beta R^2} \right] U_{cz}f = \frac{\eta_t^s}{U_c} U_{cz}f,$$

using (10). Thus

$$\begin{aligned} c: & f - \psi U_{cl} \frac{l}{\theta} + \eta f \Gamma_c + \frac{\eta_t^s}{U_c} U_{cc}f = \phi U_c, \\ q: & (1 + \nu_t)f - \psi U_{ql} \frac{l}{\theta} + \eta f \Gamma_q + \frac{\eta_t^s}{U_c} U_{cq}f = \phi U_q, \\ l: & -\theta f - \phi U_l + \eta f \Gamma_l + \frac{\eta_t^s}{U_c} U_{cl}f = \frac{\psi}{\theta} (U_{ll}l + U_l), \\ u: & \dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi, \\ w: & \frac{1}{R} V_{t+1,w} f = \phi \beta, \quad w_2: \quad \frac{1}{R} V_{t+1,w_2} f = -\psi \beta, \\ S: & \mathbb{E}_0 \left[-\frac{\psi}{f} U_{Sl} \frac{l}{\theta} - \frac{\phi}{f} U_S + \eta \Gamma_S + \frac{\eta_t^s}{U_c} U_{cS} \right] = \mu_t, \end{aligned} \tag{19}$$

where $\nu_t = \mu_t S_q + R^{-1} V_{t+1,Q}$. The pollution condition averages component contributions over all date- t histories. At the constrained allocation, the quotient rule gives $\Gamma_z = -U_{lz}/(\theta U_c) - (1 - \bar{\tau}_t^y) U_{cz}/U_c$, hence

$$\begin{aligned} \Gamma_c &= (1 - \bar{\tau}_t^y) \frac{\sigma}{c} - \frac{\gamma_c}{y}, \\ \Gamma_q &= (1 - \bar{\tau}_t^y) \frac{\sigma_{cq}}{q} - \frac{\gamma_q(1 + \tau^q)}{y}, \\ \Gamma_l &= (1 - \bar{\tau}_t^y) \frac{\epsilon - \gamma_c}{l}, \\ \Gamma_S &= -\frac{U_S}{U_c} \left[\frac{\xi}{y} + (1 - \bar{\tau}_t^y) \frac{\xi_c}{c} \right]. \end{aligned} \tag{20}$$

Here the cross-curvature convention is that of (2). The factor $1 + \tau^q$ in Γ_q follows from $U_{ql}l = \gamma_q U_q$. Appendix B.2 gives the additional hours derivative when the labor schedule is indexed by earnings.

Commodity wedge and retirement. Eliminating ϕ between the consumption and dirty-good conditions in (19) gives

$$\frac{\tau_t^q - \nu_t}{1 + \tau_t^q} = (\gamma_{c,t} - \gamma_{q,t}) \frac{\psi U_{c,t}}{\theta f_t} - \eta_t \Gamma_{c,t} + \eta_t^s \frac{\sigma_t}{c_t} + \frac{\eta_t \Gamma_{q,t} - \eta_t^s \sigma_{cq,t}/q_t}{1 + \tau_t^q}.$$

Using (20) and $\mathcal{I}_{c,t}^q = \sigma_t/c_t - \sigma_{cq,t}/[q_t(1 + \tau_t^q)]$, collecting the two multipliers yields

$$\frac{\tau_t^q - \nu_t}{1 + \tau_t^q} = (\gamma_{c,t} - \gamma_{q,t}) \frac{\psi U_{c,t}}{\theta f_t} - \eta_t \mathcal{I}_{y,t}^q - \zeta_t \mathcal{I}_{c,t}^q,$$

where

$$\mathcal{I}_{y,t}^q = \frac{1}{\theta_t} \left(\frac{U_{ql}}{U_q} - \frac{U_{cl}}{U_c} \right) = -\frac{\gamma_{c,t} - \gamma_{q,t}}{y_t}.$$

Appendix B.4 identifies $\nu_t = \text{SCC}_t$, completing Proposition 5.1.

At fixed labor and pollution, the demand identity (37) shows that $\mathcal{I}_c^q$ has the sign of the dirty good's response to expenditure.

Corollary B.1 (Retirement).

From T_R on, the current labor-wedge constraint and reporting motive are absent, so $\eta_t = \psi = 0$ and $\zeta_t = -\eta_t^s$. Without requiring weak separability,

$$\frac{\tau_t^q - \text{SCC}_t}{1 + \tau_t^q} = \eta_t^s \mathcal{I}_{c,t}^q.$$

Proof. Set the labor and reporting terms to zero in the goods conditions of (19). $\square$

B.2 Labor multiplier and carbon-price decomposition

This subsection derives the labor multiplier and an alternative carbon-price decomposition in terms of the labor gap. All objects are evaluated at one constrained allocation; the decomposition does not determine the savings multiplier or identify the effect of changing its restriction.

Conditional labor responses. Define

$$\begin{aligned} \Gamma_{c,t} &= (1 - \bar{\tau}_t^y) \frac{\sigma_t}{c_t} - \frac{\gamma_{c,t}}{y_t}, & \Lambda_t &= \frac{\Gamma_{c,t}}{1 - \bar{\tau}_t^y}, \\ \Delta_t &= (1 - \bar{\tau}_t^y) \frac{\sigma_t}{c_t} + \frac{\epsilon_t - 2\gamma_{c,t}}{y_t}. \end{aligned} \tag{21}$$

All responses below hold dirty-good consumption and resources at other dates fixed. Write $b_t = 1 - \bar{\tau}_t^y > 0$. The identity $\theta b_t \Gamma_{c,t} + \Gamma_{l,t} = \theta b_t \Delta_t$ turns the differentiated hours condition, with $dc = \theta b_t dl + dm$, into

$$\frac{\partial l}{\partial m} = -\frac{\Gamma_{c,t}}{\theta b_t \Delta_t}.$$

Under utility compensation, $dc = \theta b_t dl$, and changing the net-of-wedge rate gives

$$s_{yy,t} \equiv \left. \frac{\partial y}{\partial b_t} \right|_{u,q} = \frac{1}{b_t \Delta_t}. \quad (22)$$

The same statistic determines the household's second-order condition along the budget $c = \theta b_t l + m$:

$$\frac{d^2 U}{dl^2} = \theta^2 b_t^2 U_{cc} + 2\theta b_t U_{cl} + U_{ll} = -\theta^2 b_t U_c \Delta_t.$$

At an interior stationary point, $\Delta_t > 0$ is the strict conditional second-order condition, and $\Gamma_{c,t} > 0$ then gives normal leisure. These are the maintained restrictions. They do not sign the total labor response when dirty-good demand and saving also adjust, and neither weak separability nor Assumption 3.1 alone guarantees $\Delta_t > 0$. An earnings-indexed schedule replaces Δ_t with $\Delta_t + p_t/y_t$.

The labor multiplier.

Lemma B.1 (The labor multiplier).

Let $A_t = 1 + \epsilon_t - \gamma_{c,t}$ and define the desired labor-wedge ratio and the gap at the constrained allocation by

$$\frac{\tau_t^{y,*}}{1 - \tau_t^{y,*}} = A_t \frac{\psi U_{c,t}}{\theta f_t}, \quad \mathcal{E}_t = \frac{\bar{\tau}_t^y}{1 - \bar{\tau}_t^y} - \frac{\tau_t^{y,*}}{1 - \tau_t^{y,*}}.$$

At a working-life history,

$$\eta_t = \frac{\mathcal{E}_t + \eta_t^s \Lambda_t}{\Delta_t}. \quad (23)$$

Thus η_t has the sign of the gap adjusted for the savings multiplier.

Proof. The consumption condition in (19) gives

$$\frac{U_c \phi}{f} = 1 - \gamma_c \frac{\psi U_c}{\theta f} + \eta \Gamma_c - \eta_t^s \frac{\sigma}{c} \equiv 1 - \chi_t. \quad (24)$$

Eliminate ϕ between the consumption and labor conditions, using $U_l = -\theta U_c(1 - \bar{\tau}_t^y)$ and $U_{ll}l + U_l = (1 + \epsilon)U_l$. The labor-multiplier coefficient becomes $\Gamma_c + \Gamma_l/[\theta(1 - \bar{\tau}_t^y)] = \Delta_t$; the savings coefficient is $-\sigma/c + \gamma_c/[(1 - \bar{\tau}_t^y)y] = -\Lambda_t$. Therefore

$$\frac{\bar{\tau}_t^y}{1 - \bar{\tau}_t^y} = A_t \frac{\psi U_{c,t}}{\theta f_t} + \eta_t \Delta_t - \eta_t^s \Lambda_t, \quad (25)$$

which gives (23). □

The desired wedge is evaluated at this allocation. Re-solving after lifting a restriction also changes its costate and allocation, so the desired wedge need not equal the wedge in that new solution.

The carbon-price decomposition.

Lemma B.2 (Decomposition of the net shadow price).

At a working-life history,

$$\zeta_t = \frac{1 - \bar{\tau}_t^y}{\Delta_t} \mathcal{E}_t - \frac{A_t - 1}{y_t \Delta_t} \eta_t^s. \quad (26)$$

The labor-gap coefficient is positive. If $A_t > 1$ throughout working life, the savings-multiplier coefficient is negative at every working date; in retirement it is -1 . Its contribution to ζ_t therefore takes both signs whenever η_t^s does.

Proof. Substitute (23) into $\zeta_t = (1 - \bar{\tau}_t^y)\eta_t - \eta_t^s$. Since $(1 - \bar{\tau}_t^y)\Lambda_t = \Gamma_{c,t}$ and $\Gamma_{c,t} - \Delta_t = -(A_t - 1)/y_t$, the savings coefficient is $-(A_t - 1)/(y_t \Delta_t)$. Positivity of Δ_t and $1 - \bar{\tau}_t^y$ gives the signs. Corollary B.1 gives the retirement coefficient. $\square$

Under weak separability, write $U(c, q, l, S) = F(h(c, q, S), l, S)$. Then

$$\frac{U_q}{U_c} = \frac{h_q}{h_c}, \quad \gamma_c = \gamma_q = \frac{l F_{hl}}{F_h}.$$

The common complementarity need not vanish; additive separability between consumption and labor sets $F_{hl} = 0$. Combining (26) with Corollary 5.1 therefore gives (27) while retaining $\gamma_{c,t}$.

$$\frac{\tau_t^q - \text{SCC}_t}{1 + \tau_t^q} = \mathcal{I}_{c,t}^q \left[-\frac{1 - \bar{\tau}_t^y}{\Delta_t} \mathcal{E}_t + \frac{\epsilon_t - \gamma_{c,t}}{y_t \Delta_t} \eta_t^s \right], \quad t < T_R. \quad (27)$$

For a normal dirty good, the term proportional to η_t^s in (27) has its sign if $A_t > 1$, or equivalently $\epsilon_t > \gamma_{c,t}$. This is an additional restriction, stronger than Assumption 3.1's $A_t > 0$ and not implied by weak separability. Holding both goods and pollution fixed,

$$\frac{\partial \ln(-U_l/U_c)}{\partial \ln l} = \epsilon - \gamma_c.$$

The restriction says that the net wage required to induce more labor rises with hours at a fixed consumption bundle. Here ϵ measures labor curvature holding consumption fixed; with consumption-labor interactions, an inverse Frisch elasticity also involves cross derivatives because marginal utility of wealth is held fixed.

The labor-gap and savings-multiplier terms are components of the same first-order condition. Neither is a separate policy effect. Even when labor is unconstrained, $\eta_t = 0$ implies $\mathcal{E}_t = -\eta_t^s \Lambda_t$, so the labor-gap term can be nonzero. In that case the two terms sum to $\eta_t^s \mathcal{I}_{c,t}^q$. Re-solving after changing a fixed schedule changes both terms and the allocation.

Earnings-indexed labor schedules. The tax function of Bénabou (2002) and Heathcote et al. (2017) is

$$T_t(y) = y - \lambda_t^T y^{1-p_t}, \quad 1 - \bar{\tau}_t^y(y) = \lambda_t^T (1 - p_t) y^{-p_t}. \quad (28)$$

Since $\partial_y \bar{\tau}_t^y = p_t(1 - \bar{\tau}_t^y)/y$, differentiating the labor constraint changes only its hours derivative:

$$\Gamma_l = (1 - \bar{\tau}_t^y) \frac{\epsilon - \gamma_c + p_t}{l}.$$

The other derivatives in (20) are unchanged. For the compensated labor response, the change in the marginal net-of-tax rate is measured at the original earnings level, holding progressivity fixed. Consequently the labor formula and conditional response in (25) and (22) use $\Delta_t + p_t/y_t$, with Δ_t retaining its type-indexed definition in (21).

In the net-shadow-price identity, the numerator changes too: $\Gamma_{c,t} - \Delta_t - p_t/y_t = -(A_t - 1 + p_t)/y_t$. Thus

$$\zeta_t = \frac{(1 - \bar{\tau}_t^y) \mathcal{E}_t}{\Delta_t + p_t/y_t} - \frac{A_t - 1 + p_t}{y_t \Delta_t + p_t} \eta_t^s.$$

The savings coefficient is negative when $A_t - 1 + p_t > 0$. Progressivity also strengthens the household's conditional second-order condition by adding p_t/y_t to Δ_t .

In the carbon-price decomposition (27), replace Δ_t by $\Delta_t + p_t/y_t$ and $\epsilon_t - \gamma_{c,t}$ by $\epsilon_t - \gamma_{c,t} + p_t$. The corresponding strict second-order condition and sign restriction are $\Delta_t + p_t/y_t > 0$ and $\epsilon_t - \gamma_{c,t} + p_t > 0$.

B.3 Savings multiplier over the life cycle

Sign reversal. Fix a realized history. Let $\kappa_t = (1 - \bar{\tau}_t^s) \beta R^2 > 0$, $P_t = \prod_{j < t} \kappa_j$, with $P_0 = 1$, and $x_t = \vartheta_t/P_t$. Equation (10) gives

$$x_{t+1} = x_t - \frac{\eta_t^s}{P_t U_{c,t}}, \quad t < T. \quad (29)$$

The endpoint conditions $\vartheta_0 = \vartheta_{T+1} = 0$ imply $x_0 = 0$ and $x_T = \eta_T^s/(P_T U_{c,T})$. Summing (29),

$$\sum_{t=0}^T \frac{\eta_t^s}{P_t U_{c,t}} = 0 \quad (30)$$

If any multiplier along the history is nonzero, terms of both signs must appear.

The intertemporal condition. The unchanged promise and costate conditions give

$$\lambda_{1,t+1} = \beta R \frac{1 - \chi_t}{U_{c,t}}, \quad \lambda_{1,t} = \mathbb{E}_{t-1} \left[\frac{1 - \chi_t}{U_{c,t}} \right],$$

where the second equality integrates the costate equation using $\psi(0) = \psi(\infty) = 0$. Equating them at adjacent dates yields

$$\frac{1 - \chi_t}{U_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1 - \chi_{t+1}}{U_{c,t+1}} \right]. \quad (31)$$

Separate the explicit savings term in (24):

$$\mathcal{G}_t = 1 - \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} + \eta_t \Gamma_{c,t}, \quad 1 - \chi_t = \mathcal{G}_t - \eta_t^s \frac{\sigma_t}{c_t}. \quad (32)$$

$\mathcal{G}_t$ is dimensionless and retains the savings-wedge constraint's indirect effects through η_t , the costate, and the allocation. Where $\mathcal{G}_t > 0$ and $\mathbb{E}_t[\mathcal{G}_{t+1}] > 0$, define the reference wedge by $1 - \tau_t^{s,*} = \mathcal{G}_t / \mathbb{E}_t[\mathcal{G}_{t+1}]$.

Multiplying (31) by $U_{c,t}$ and substituting the Euler equation with the prescribed savings wedge gives

$$1 - \chi_t = (1 - \bar{\tau}_t^s) \left\{ \mathbb{E}_t[1 - \chi_{t+1}] - \text{Cov}_t \left(U_{c,t+1}, \frac{1 - \chi_{t+1}}{U_{c,t+1}} \right) \right\}.$$

Substitution of (32) then yields

$$\begin{aligned} \eta_t^s \frac{\sigma_t}{c_t} - (1 - \bar{\tau}_t^s) \mathbb{E}_t \left[\eta_{t+1}^s \frac{\sigma_{t+1}}{c_{t+1}} \right] &= \mathbb{E}_t[\mathcal{G}_{t+1}] (\bar{\tau}_t^s - \tau_t^{s,*}) \\ &+ (1 - \bar{\tau}_t^s) \text{Cov}_t \left(U_{c,t+1}, \frac{1 - \chi_{t+1}}{U_{c,t+1}} \right). \end{aligned} \quad (33)$$

When the next draw is degenerate this reduces to

$$\eta_t^s \frac{\sigma_t}{c_t} - (1 - \bar{\tau}_t^s) \eta_{t+1}^s \frac{\sigma_{t+1}}{c_{t+1}} = \mathcal{G}_{t+1} (\bar{\tau}_t^s - \tau_t^{s,*}). \quad (34)$$

The reference wedge is therefore a benchmark at the constrained allocation, not the result of a separate unconstrained solve. Under risk, the wedge gap alone does not sign the right side of (33).

In retirement the labor-wedge constraint and costate are absent, giving $\mathcal{G}_t = 1$ directly, without evaluating Γ_c at zero earnings. For $T_R \leq t < T$, (34) becomes

$$\eta_t^s \frac{\sigma_t}{c_t} - (1 - \bar{\tau}_t^s) \eta_{t+1}^s \frac{\sigma_{t+1}}{c_{t+1}} = \bar{\tau}_t^s.$$

A nonzero fixed wedge on such an interval rules out zero savings multipliers at both adjacent dates.

Direction of the reversal. For a deterministic history, let $z_t = \eta_t^s \sigma_t / c_t$ and assume $\sigma_t > 0$, $\mathcal{G}_t > 0$, and $1 - \bar{\tau}_t^s > 0$. Strict curvature is needed because Assumption 3.1 alone permits $\sigma_t = 0$. Equation (34) reads

$$z_t = a_t z_{t+1} + g_t, \quad a_t = 1 - \bar{\tau}_t^s > 0, \quad g_t = \mathcal{G}_{t+1}(\bar{\tau}_t^s - \tau_t^{s,*}).$$

If every $g_t > 0$, then $z_t \leq 0$ implies $z_{t+1} < 0$. Some multiplier must be nonzero, and (30) supplies both signs. The sequence therefore starts positive and ends negative, crossing once, possibly through one zero date. If every $g_t < 0$, the ordering reverses. This proves the deterministic ordering stated in Section 5.1.1.

Under uncertainty, (33) contains a conditional expectation and a covariance. A positive wedge gap does not sign the total drift, and a signed conditional mean does not sign every successor. The deterministic ordering therefore does not extend to realized stochastic paths. The pathwise sign reversal still holds whenever a multiplier is nonzero. A sufficient condition is that each history has a nonzero fixed wedge on an interval between retirement dates; it requires $T_R < T$.

Population mean marginal utility. If the savings wedge is common across the date- t population, unconditional expectations in (9) give

$$\mathbb{E}_0[U_{c,t}] = \left\{ \prod_{j < t} \frac{1}{(1 - \bar{\tau}_j^s) \beta R} \right\} \mathbb{E}_0[U_{c,0}].$$

The schedule fixes growth, while the allocation problem determines the initial level. At $\beta R = 1$ and $\bar{\tau}^s = 0.009$, mean marginal utility rises by about 70% over 59 transitions.

B.4 The constrained social cost of carbon

Substitute (24) and (20) into the pollution-stock condition in (19). Using $U_{Sl} = \xi U_S$ and $U_{cS}/U_c = \xi_c(U_S/U_c)/c$, its integrand collects as

$$-\frac{U_S}{U_c} \left[1 + (\xi - \gamma_c) \frac{\psi U_c}{\theta f} + \eta \frac{\xi - \gamma_c}{y} + ((1 - \bar{\tau}_t^y) \eta - \eta_t^s) \frac{\sigma + \xi_c}{c} \right].$$

Using the willingness-to-pay gradients in (4) and averaging over all date- t histories gives

$$\mu_t = -\mathbb{E}_0 \left[\left(1 + (\xi - \gamma_c) \frac{\psi U_c}{\theta f} + \eta \mathcal{I}_y^S + \zeta_t \mathcal{I}_c^S \right) \frac{U_S}{U_c} \right]. \quad (35)$$

The emission-price recursion is unchanged. With $\Phi_j = \partial S_{t+j} / \partial Q_t$ and $\Phi_j = 0$ for $j > s$,

$$\nu_t = \text{SCC}_t = \sum_{j=0}^s R^{-j} \mu_{t+j} \Phi_j.$$

These are population multipliers and aggregate pollution responses; aggregate paths are deterministic in the model. Substituting (35) gives the damage weight in Proposition 5.1. At retirement dates the bracket reduces directly to $1 - \eta_t^s \mathcal{I}_c^S$, because the costate and labor-wedge constraint are absent. Under $U = V(h(c, q, S), l)$, $\xi = \gamma_c = \gamma_q$ removes the screening term, leaving $1 + \zeta_t \mathcal{I}_c^S$.

B.5 Carbon-only instruments and implementation

Fix differentiable income tax functions $T_t^y(y)$ and a constant tax τ^k on interest income, giving the gross return $\tilde{R} = 1 + (R - 1)(1 - \tau^k) > 0$. A carbon-only reform changes taxes $\mathcal{T}_t(q^t)$ that depend only on the date and dirty-good purchase history. Taxes may be nonlinear or negative, allowing transfers conditioned on purchases, but they cannot condition on earnings, assets, or clean-good consumption.

A household maximizes $\mathbb{E}_0[\sum_{t=0}^T \beta^t U(c_t, q_t, l_t, S_t)]$ subject to

$$c_t + q_t + \mathcal{T}_t(q^t) + a_{t+1} = \theta_t l_t - T_t^y(\theta_t l_t) + \tilde{R} a_t, \quad a_0 = a_{T+1} = 0.$$

Government accounts and asset markets clear with the same aggregate initial capital K_0 , pollution law, and terminal resource conditions as the allocation problem. Since households start with $a_0 = 0$, initial capital is held publicly.

Plans are measurable in skill history, $l_t = 0$ from T_R on, and labor and saving are interior when they are choices. With budget multiplier μ_t , the first-order conditions are

$$\begin{aligned} U_{c,t} &= \mu_t, \\ -U_{l,t} &= \mu_t \theta_t [1 - T_t^{y'}(y_t)], \quad t < T_R, \\ \mu_t &= \beta \tilde{R} \mathbb{E}_t[\mu_{t+1}], \quad t < T, \\ U_{q,t} &= \mu_t \left(1 + \frac{\partial \mathcal{T}_t}{\partial q_t} \right) + \sum_{j=1}^{T-t} \beta^j \mathbb{E}_t \left[\mu_{t+j} \frac{\partial \mathcal{T}_{t+j}}{\partial q_t} \right]. \end{aligned}$$

Tax derivatives enter only the commodity condition because the tax takes purchases, not hours or assets, as its arguments. Substituting $\mu_t = U_{c,t}$ gives the labor- and savings-wedge constraints at $\bar{\tau}_t^y(y) = T_t^{y'}(y)$ and $1 - \bar{\tau}^s = \tilde{R}/R$.

Every such equilibrium also satisfies global incentive compatibility: a household can imitate any nonanticipating reporting strategy's earnings, purchases, and asset plan under the same taxes. Under the maintained regularity conditions, global incentive compatibility implies the local envelope conditions. Together with aggregate feasibility and the income-wedge restrictions, this places attainable allocations in the relaxed allocation problem, provided both problems use the same income schedules. Its global optimum therefore bounds attainable welfare from

above. This inclusion does not establish that a numerical solution attains the global optimum or that every relaxed allocation is implementable.

The restrictions can rule out the full optimum. If that allocation has a positive savings wedge at a working-life history and a zero wedge at a retirement date with saving, no constant $\bar{\tau}^s$ matches both. Likewise, implementation fails wherever its labor wedge differs from the prescribed $T_t^{y'}(y_t)$. Additive separability between consumption and labor supplies a sufficient case for the positive working-life wedge: the inverse Euler equation gives

$$1 - \tau_t^s = \frac{1}{\mathbb{E}_t[U_{c,t+1}] \mathbb{E}_t[1/U_{c,t+1}]} \leq 1,$$

strictly below one when marginal utility is nondegenerate (Goloso et al. 2003). Weak separability alone does not remove the incentive credit in (18); under the paper's nonseparable preferences the positive wedge is a property of the numerical solution. Setting $\tau^k = 0$ still imposes the household Euler equation. State-contingent wealth taxes can change that restriction (Kocherlakota 2005).

A candidate tax and its limitation. Given a constrained allocation, set its asset path to zero and assign the on-path taxes

$$\mathcal{T}_t(q^t(\theta^t)) = \theta_t l_t(\theta^t) - T_t^y(\theta_t l_t(\theta^t)) - c_t(\theta^t) - q_t(\theta^t).$$

This construction requires purchase histories to separate any skill histories assigned different taxes. Any extension of the tax outside the assigned purchase histories leaves deviations to other assigned histories available.

Holding a purchase plan fixed, the household can still choose hours, clean consumption, and saving. Joint concavity of U and concavity of after-tax earnings make the remaining choice problem concave, so the income-wedge conditions certify its truthful bundle. Joint concavity is an additional condition, not part of Assumption 3.1. Let $J(\theta, r)$ be the direct-mechanism payoff from the bundle assigned to report r , and let $W(\theta, r)$ allow these remaining choices to be re-optimized at tax position r . The assigned bundle is feasible there, hence $W(\theta, r) \geq J(\theta, r)$, with equality at truth under the concavity conditions. A maximum of $J(\theta, \cdot)$ need not maximize $W(\theta, \cdot)$.

Counterexample. Take work at date 0 and retirement at date 1, with $U = \log c + \log q - l^2/2$, $\beta R = 1$, both income taxes zero, and $K_0 = 0$. Initial skill is continuously distributed on a sufficiently small neighborhood of $\theta^\circ = 1/[2(1+\beta)]$. The pollution term is common to all plans and can be omitted. Consider

$$c_0 = c_1 = \theta^{3/2}, \quad y_0 = \theta^{1/2}, \quad q_1 = d, \quad q_0 = C\theta^{-3(1+\beta)/2} \exp\left(-\frac{1}{2\theta}\right),$$

Choose $C, d > 0$ to satisfy the aggregate resource constraint:

$$\int [q_0(\theta) + \beta d] dF_0(\theta) = \int \sqrt{\theta} [1 - (1 + \beta)\theta] dF_0(\theta).$$

The right-hand side is positive on this skill range, so such constants exist. At the assigned allocation, the candidate tax has a balanced present-value government budget: date-0 revenue is saved at return R to fund retirement transfers. The labor and Euler conditions with fixed income wedges hold: $y_0 c_0 / \theta^2 = 1$ and $c_0 = c_1$. A type θ reporting r obtains, up to constants,

$$J(\theta, r) = -\frac{1}{2r} - \frac{r}{2\theta^2}, \quad J_r(\theta, r) = \frac{1}{2r^2} - \frac{1}{2\theta^2}.$$

This derivative is positive below truth and negative above it, establishing global incentive compatibility of the direct mechanism. Purchases separate types near θ° because $q'_0(r)/q_0(r) = -3(1 + \beta)/(2r) + 1/(2r^2) < 0$ there.

Under the candidate tax, position r supplies lifetime resources $B(r) = (1 + \beta)r^{3/2} - r^{1/2}$ for clean consumption and labor. Re-optimization gives $c_0 = c_1 = c$ with $(1 + \beta)c = y + B(r)$ and $yc = \theta^2$, so

$$W_r(\theta, r) = \frac{B'(r)}{c(\theta, r)} + \frac{q'_0(r)}{q_0(r)}.$$

At truth $W_r(\theta, \theta) = 0$. Differentiating this identity gives $W_{rr}(\theta, \theta) = -W_{r\theta}(\theta, \theta)$, while $W_{r\theta} = -B'(r)c_\theta/c^2$. At truth, $c = \theta^{3/2}$ and $c_\theta = 2\sqrt{\theta}/[1 + (1 + \beta)\theta]$, giving

$$W_{rr}(\theta, \theta) = \frac{3(1 + \beta)\theta - 1}{[1 + (1 + \beta)\theta]\theta^3}, \quad W_{rr}(\theta^\circ, \theta^\circ) = \frac{1}{3(\theta^\circ)^3} > 0.$$

Truth is a strict local minimum of the tax payoff for types near θ° . Neighboring menu positions are profitable deviations that off-menu penalties cannot remove, even though joint concavity and the income-wedge conditions hold.

The example does not show that the computed allocation fails: its purchases decline with skill and it is not a planner optimum for any stated welfare weights. It establishes that the stated conditions do not suffice. The remaining check is household re-optimization under the candidate tax, including hours and assets.

C Derivations: a common carbon price

This appendix proves the common-price results of Section 5.2. It first establishes the expenditure reduction and substitution weights, then derives the optimal rate with income wedges free or fixed. Every statistic and desired deviation is evaluated at the allocation satisfying the price constraint. The substitution responses and weighted corrections below are assumed integrable over histories, so the averaging weights have finite, positive totals.

At date t , impose the common consumer price $\bar{p}_t = 1 + \bar{\tau}_t^q > 0$ through

$$\Gamma^q \equiv U_q/U_c - \bar{p}_t = 0. \quad (36)$$

C.1 Expenditure, demand, and incentive compatibility

At fixed price, labor, and pollution, define indirect utility by

$$\tilde{U}(x, l; \bar{p}, S) = \max_{c, q \geq 0} U(c, q, l, S) \quad \text{s.t.} \quad c + \bar{p}q = x.$$

Lemma C.1 (Common pricing and the expenditure reduction).

Under Assumption 3.1, an interior allocation satisfies (36) if and only if it assigns each type the unique private split of some (x, y) at the common price. The local first-order problem can therefore be written in (x, y) with period utility $\tilde{U}(x, y/\theta; \bar{p}_t, S_t)$ and truthful envelope

$$\dot{u}(\theta) = -\tilde{U}_l l/\theta + \beta w_2(\theta) = -U_l l/\theta + \beta w_2(\theta).$$

Proof. Substituting $c = x - \bar{p}q$ gives the first-order condition $U_q = \bar{p}U_c$. The regular strict quasi-concavity and Inada conditions in Assumption 3.1 give a unique interior split. At that split,

$$\mathcal{D} \equiv U_{qq} - 2\bar{p}U_{cq} + \bar{p}^2 U_{cc} < 0.$$

The envelope theorem gives $\tilde{U}_x = U_c$ and $\tilde{U}_l = U_l$. True skill enters period utility through $l = y/\theta$, so the truthful envelope is unchanged. Promise keeping and resource costs can also be expressed in (x, y) , since $c + q = x - \bar{\tau}^q q^d(x, l; \bar{p}, S)$. The same identities preserve the income-wedge restrictions:

$$-\frac{\tilde{U}_{l,t}}{\theta_t \tilde{U}_{x,t}} = 1 - \bar{\tau}_t^y, \quad \tilde{U}_{x,t} = (1 - \bar{\tau}_t^s) \beta R \mathbb{E}_t[\tilde{U}_{x,t+1}],$$

with the labor condition imposed before retirement and the savings condition before the terminal date. $\square$

Global deviations require a separate check. A type θ imitating $\hat{\theta}$ can re-split expenditure at labor $y(\hat{\theta})/\theta$, giving

$$\tilde{U}\left(x(\hat{\theta}), \frac{y(\hat{\theta})}{\theta}; \bar{p}, S\right) \geq U\left(c(\hat{\theta}), q(\hat{\theta}), \frac{y(\hat{\theta})}{\theta}, S\right).$$

Equality holds at truth and, under weak separability from labor, at every report. Otherwise the assigned-bundle payoff can understate the deviation gain.

Demand identities. Differentiate $U_q = \bar{p}U_c$ jointly with the expenditure constraint. At fixed labor and pollution, expenditure and compensated price changes give

$$\begin{aligned} \partial_x q^d &= \frac{U_{cq} - \bar{p}U_{cc}}{|\mathcal{D}|} = \frac{\bar{p}U_c \mathcal{I}_c^q}{|\mathcal{D}|}, \\ s_{qq} &\equiv \frac{\partial q^d}{\partial \bar{p}} \Big|_u = \frac{U_c}{\mathcal{D}} < 0. \end{aligned} \tag{37}$$

The second identity uses $dc = -\bar{p}dq$ along an indifference curve. The first shows that $\mathcal{I}_c^q$ is positive exactly when the dirty good is normal. The combination of constraint gradients needed below is

$$\Delta_t^q \equiv \bar{p}\mathcal{I}_c^q - \mathcal{I}_q^q = -\frac{\mathcal{D}}{\bar{p}U_c} = \frac{1}{\bar{p}|s_{qq}|} > 0. \tag{38}$$

C.2 The optimal common price

First-order conditions. The Hamiltonian of Appendix A.1 gains $\eta^q f \Gamma^q$. Extending the log-gradient definition of $\mathcal{I}_z^q$ to any utility argument z , the quotient rule at a fixed prescribed price gives

$$\Gamma_z^q = \frac{U_{qz} - \bar{p}U_{cz}}{U_c} = \bar{p}\mathcal{I}_z^q. \tag{39}$$

Each pointwise condition for c, q , and l gains $\eta^q f \Gamma_z^q$; the costate and continuation conditions are unchanged. In particular, with income wedges free, the conditions for the two goods are

$$\frac{\phi U_c}{f} = 1 - \gamma_c \frac{\psi U_c}{\theta f} + \eta^q \bar{p} \mathcal{I}_c^q, \tag{40}$$

$$\frac{\phi U_c}{f} = \frac{1 + \nu_t}{\bar{p}} - \gamma_q \frac{\psi U_c}{\theta f} + \eta^q \mathcal{I}_q^q. \tag{41}$$

Here ν_t is the shadow emission price in (13). The direct addition to the aggregate pollution-stock condition is $\mathbb{E}_0[\eta^q \Gamma_S^q]$; Appendix C.4 combines it with the change in the utility-delivery cost to derive the social cost of carbon.

The optimal rate. Equating (40) and (41) gives

$$\underbrace{1 - \frac{1 + \nu_t}{\bar{p}_t}}_{\varpi_t} = \underbrace{(\gamma_c - \gamma_q) \frac{\psi U_c}{\theta f}}_{\varpi_t^*} - \eta_t^q \Delta_t^q.$$

Using (38), the multiplier is the desired price gap converted into resources by the substitution response:

$$\eta_t^q = \bar{p}_t |s_{qq,t}| (\varpi_t^* - \varpi_t). \quad (42)$$

The common rate enters the date-0 resource-cost Lagrangian only through the date- t price constraints. Accounting for resource discounting and transition probabilities gives

$$\frac{d\mathcal{L}_0}{d\bar{\tau}_t^q} = -R^{-t} \mathbb{E}_0[\eta_t^q]. \quad (43)$$

The planner minimizes resource cost at fixed promised welfare, so a negative derivative favors raising the rate. An interior optimum sets this derivative to zero. Since $\bar{p}_t$ and ϖ_t are common across date- t histories, substitution into (42) proves Proposition 5.2:

$$\varpi_t = \frac{\mathbb{E}_0[|s_{qq,t}| \varpi_t^*]}{\mathbb{E}_0[|s_{qq,t}|]}.$$

Weak separability gives $\varpi_t^* = 0$ at every history; retirement gives the same result through $\psi = 0$. In either case the optimal date-specific rate equals ν_t . Under weak separability at every date, the unrestricted optimum already satisfies the common-price restriction, establishing the no-loss claim. These identities are evaluated at the constrained allocation and do not average rates from a separately solved history-dependent optimum.

C.3 Fixed income wedges and a lifetime rate

Corollary C.1 (The optimal common price against fixed income wedges).

Add the common-price constraint to the problem of Section 5.1, with its level chosen at each date. At the joint constrained allocation, define

$$\varpi_t^{**} = \begin{cases} \varpi_t^* + (\gamma_{c,t} - \gamma_{q,t}) \eta_t / y_t - \zeta_t \mathcal{I}_{c,t}^q, & t < T_R, \\ \eta_t^s \mathcal{I}_{c,t}^q, & t \geq T_R. \end{cases}$$

Then $\eta_t^q = \bar{p}_t |s_{qq,t}| (\varpi_t^{**} - \varpi_t)$, and an interior optimal common price satisfies

$$\varpi_t = \frac{\mathbb{E}_0[|s_{qq,t}| \varpi_t^{**}]}{\mathbb{E}_0[|s_{qq,t}|]}.$$

The social cost of carbon uses the damage weight for fixed income wedges, augmented by the price-constraint term derived in Appendix C.4.

Proof. The pointwise conditions for c, q , and l in (19) gain $\eta^q f \Gamma_z^q$. Equating the resulting conditions for c and q , as above, gives

$$\varpi_t = \varpi_t^* - \eta_t \left(\Gamma_c - \frac{\Gamma_q}{\bar{p}_t} \right) + \eta_t^s \mathcal{I}_{c,t}^q - \eta_t^q \Delta_t^q.$$

At working-life histories, (20) implies

$$\Gamma_c - \frac{\Gamma_q}{\bar{p}_t} = (1 - \bar{\tau}_t^y) \mathcal{I}_{c,t}^q - \frac{\gamma_{c,t} - \gamma_{q,t}}{y_t}.$$

Substitution and $\zeta_t = \eta_t(1 - \bar{\tau}_t^y) - \eta_t^s$ give $\varpi_t = \varpi_t^{**} - \eta_t^q \Delta_t^q$. In retirement, screening and the labor constraint are absent, leaving $\varpi_t^{**} = \eta_t^s \mathcal{I}_{c,t}^q$. Equation (38) yields the multiplier formula. Neither income-wedge constraint directly contains the prescribed carbon-price level, so (43) still applies and proves the average. The income multipliers belong to this joint problem; their values need not equal those obtained with the carbon wedge free. $\square$

In retirement, the common price equals the social cost of carbon exactly when $\mathbb{E}_0[|s_{qq,t}| \eta_t^s \mathcal{I}_{c,t}^q] = 0$. Vanishing savings multipliers suffice, but cancellation across histories can also deliver equality. Weak separability alone removes neither this term nor the working-life correction for fixed income wedges.

Corollary C.2 (The linear rate against fixed income wedges).

Restrict Corollary C.1 to one rate at every date, $\bar{\tau}_t^q = \bar{\tau}^q$. An interior optimum satisfies

$$\sum_{t=0}^T R^{-t} \mathbb{E}_0[\eta_t^q] = 0.$$

When the social cost of carbon is constant across dates, this gives (12). With income wedges free, replace ϖ_t^{**} by ϖ_t^* .

Proof. The lifetime rate enters every date's price constraint. Summing (43) across dates and substituting the multiplier formula gives

$$\sum_{t=0}^T R^{-t} \mathbb{E}_0[|s_{qq,t}| (\varpi_t^{**} - \varpi_t)] = 0.$$

For a time-varying social cost of carbon, $\varpi_t = (\bar{\tau}^q - \nu_t)/\bar{p}$ and this is equivalent to

$$\bar{\tau}^q = \frac{\sum_{t=0}^T R^{-t} \mathbb{E}_0[|s_{qq,t}| (\nu_t + \bar{p} \varpi_t^{**})]}{\sum_{t=0}^T R^{-t} \mathbb{E}_0[|s_{qq,t}|]}.$$

When ν_t is constant, it can be taken outside the average, proving (12). Removing the income constraints sets their multipliers to zero and replaces ϖ_t^{**} by ϖ_t^* . $\square$

The per-date multiplier means need not vanish. With a constant social cost of carbon and both working and retirement dates present, the lifetime average is a convex combination of their desired-deviation means, using total weights $\sum_{t < T_R} R^{-t} \mathbb{E}_0 |s_{qq,t}|$ and $\sum_{t \geq T_R} R^{-t} \mathbb{E}_0 |s_{qq,t}|$. A zero retirement mean therefore dilutes a nonzero working-life mean. This decomposes one solved allocation; re-optimizing after changing the horizon would also change the weights and deviations.

C.4 The social cost of carbon

The price constraint affects the damage weight through both the direct pollution derivative and the delivery cost ϕ . From (39) and (40), their combined contribution at each history is

$$\eta^q \left(\Gamma_S^q - \Gamma_c^q \frac{U_S}{U_c} \right) = -\eta^q \frac{U_S}{U_c} (\bar{p} \mathcal{I}_c^S - \mathcal{I}_q^S).$$

This follows directly from $\mathcal{I}_z^S = \partial_z \ln(-U_S/U_c)$ and the quotient rule. With income wedges free, substituting into the condition of Appendix A.6 yields

$$\mu_t = -\mathbb{E}_0 \left[\left(1 + (\xi - \gamma_c) \frac{\psi U_c}{\theta f} + \eta^q (\bar{p} \mathcal{I}_c^S - \mathcal{I}_q^S) \right) \frac{U_S}{U_c} \right].$$

The expectation averages all date- t histories. With income wedges fixed, the calculation adds the same η^q term to the bracket of (35). The emission recursion (13) is unchanged, so aggregating these marginal stock damages as in Appendix A.6 gives the social cost of carbon in either regime.

Choosing the common rate optimally does not generally remove the additional term. At a date whose price is free to adjust, $\mathbb{E}_0 \eta_t^q = 0$, so its population contribution is

$$\mathbb{E}_0[\eta_t^q \mathcal{K}_t] = \text{Cov}_0(\eta_t^q, \mathcal{K}_t), \quad \mathcal{K}_t \equiv -\frac{U_{S,t}}{U_{c,t}} (\bar{p}_t \mathcal{I}_{c,t}^S - \mathcal{I}_{q,t}^S).$$

This covariance may be nonzero. With one lifetime rate, even the per-date mean of η_t^q need not vanish. The social cost remains common across households because pollution is aggregate, but its level can change with the constrained allocation. With income wedges free and every remaining date's price chosen separately, retirement sets $\psi = \eta^q = 0$, so the damage weight is one. With income wedges free, weak separability during working life sets $\eta^q = 0$ at a freely chosen date-specific common price but does not by itself eliminate the $\xi - \gamma_c$ correction.

D Quantitative exercise

D.1 Calibrating the fixed income wedges

The labor benchmark uses the functional form in (28). Heathcote et al. (2017) estimate progressivity $p = 0.181$ for US households aged 25 to 60 in the PSID; Heathcote and Tsujiyama (2021) pair it with $\lambda^T = 0.840$. Evidence of little age dependence in progressivity supports holding p constant while allowing the level to vary (Heathcote et al. 2020).

Normalized earnings. The solver applies the schedule to $\hat{y}_t = y_t/\theta_{t-1}$:

$$1 - \bar{\tau}_t^y = \bar{\Lambda}_t \hat{y}_t^{-p} = \bar{\Lambda}_t \theta_{t-1}^p y_t^{-p}.$$

At zero progressivity the wedge is common within each date. At positive progressivity, equal observed earnings can face different wedges. For example, doubling lagged skill multiplies the net-of-tax rate by 2^p : at $p = 0.181$, a 30% tax rate becomes about 21%. Entrants share a lagged skill, so the earnings normalization is common at entry. The normalization preserves homogeneity but makes the restriction differ from a single tax schedule in observed earnings. The quantitative comparisons inherit this qualification; Section 5.1.2 discusses its implications for implementation.

The savings wedge. A linear tax τ_t^k on interest income gives the after-tax gross return $1 + (R - 1)(1 - \tau_t^k)$. The household Euler equation therefore implies

$$1 - \bar{\tau}_t^s = \frac{1 + (R - 1)(1 - \tau_t^k)}{R}. \quad (44)$$

At $\beta R = 1$, this becomes $\bar{\tau}_t^s = (1 - \beta)\tau_t^k$. The benchmark $\tau^k = 0.18$ thus fixes an annual wedge of 0.009, including retirement.

The 18% anchor is the effective marginal capital tax under 2014 law reported by Congressional Budget Office (2014). It is an approximate household-savings calibration: CBO's composite averages business capital and owner-occupied housing and includes corporate taxes, whereas the model fixes the return available to households. The 36% alternative comes from aggregate tax-revenue-to-income ratios (Mendoza et al. 1994; Trabandt and Uhlig 2011). That average rate is not an estimate of the household marginal rate. Table 4 also includes rates below the historical benchmark.

D.2 Energy calibration

The energy good combines motor fuel, electricity, natural gas, and fuel oil. Its 7% expenditure share is an approximate calibration target informed by the Consumer Expenditure Survey. The

calibration matches this share at expenditure $x = 1$, labor $l = 1$, and consumer price $\bar{p} = 1$, which pins the CES weight b_q in the homothetic benchmark. Conditional on labor, substitution elasticity 0.4 then implies a Marshallian own-price elasticity of $0.4(1 - 0.07) + 0.07 = 0.442$ in absolute value at that point. The mean long-run elasticity in Labandeira et al. (2017) is about 0.6.

The Stone-Geary variant jointly matches the same expenditure share and an income elasticity of 0.45 at the calibration point. The latter is an assumed cross-sectional target; residential-electricity estimates vary across settings and horizons (Espey and Espey 2004; Zhu et al. 2018). These targets determine b_q and $\bar{q} \approx 0.04$, with subsistence accounting for about 57% of energy spending at that point. They need not equal the corresponding moments averaged over simulated households.

The complementarity parameter $\varkappa = 0.25$ targets a mid-career energy wedge of about 10% without damages. West and Williams (2007) estimate a zero-externality gasoline tax of 33% of the pre-tax price, including 13 percentage points from nonseparability. The smaller composite-energy target is an assumption: their study has no separate home-energy good. Its estimated complementarity also changes sign when state fixed effects are omitted, although the authors reject omitting them. The scale of $\varkappa$ depends on the labor normalization because utility contains $\varkappa l^\alpha$.

The damage conversion multiplies the U.S. Environmental Protection Agency (2023) social cost of \$190 per ton, for 2020 emissions in 2020 dollars at a 2% near-term discount rate, by 0.0026 tons per producer dollar. The intensity combines 2023 household-energy emission factors and pre-tax prices; its unrounded value is nearer 0.0027. The rounding and differing dollar years make $e = 0.494$ an approximate calibration. Reported dollar prices use the fixed conversion $\tau^q \times 190/0.494$.

Consumption-unit damages give the same corrective benchmark in both income-tax regimes. Writing utility as $U(c - D(S), q, l)$ yields $U_S/U_c = -D'$ and $U_{Sl}/U_S = U_{cl}/U_c$, hence $\xi_t = \gamma_{c,t}$ in Proposition 4.2. The formula with fixed income wedges also has $\xi_{c,t} = -\sigma_t$ and $\mathcal{I}_{c,t}^S = 0$. The numerical problem incorporates this constant resource cost directly and carries no pollution-stock state.

D.3 Sensitivity to the income-wedge restrictions

Table 4 reports the sensitivity behind Section 6.2. A zero capital tax still restricts saving through the household Euler equation. Higher rates raise the entry carbon price and lower the late-life price in this grid. Crossing the social cost of carbon is not universal: the 0.025 and 0.05 paths remain below it, and the zero-rate crossing is marginal. These price crossings are distinct from the multiplier sign reversal of Lemma 5.2.

Table 4: Carbon prices across the capital-tax grid, with the labor schedule fixed throughout. The full-optimum memo row leaves both income wedges free.

τ^k	$\bar{\tau}^s$ (%/yr)	δ at 25	crossing age	δ at 64	δ at 84	working-life mean τ^q
0	0	-0.28	62	0.00	-0.07	0.38
0.025	0.13	-0.25	—	-0.08	-0.22	0.34
0.05	0.25	-0.22	—	-0.18	-0.38	0.30
0.18	0.90	+0.01	28	-0.62	-1.27	0.23
0.36	1.80	+0.29	36	-1.31	-2.98	0.22
<i>memo: full optimum</i>	free	+0.01	—	+0.08	0	0.59

Note: $\delta_t = (\tau_t^q - e)/(1 + \tau_t^q)$, with $e = 0.494$, is evaluated at the cross-sectional mean carbon wedge. This transformation is concave, so its value at the mean wedge need not equal the mean deviation. “Crossing age” is the first reported age at which δ_t changes sign; a dash denotes no crossing. The zero-rate crossing is marginal, with the deviation within about 0.003 of zero from age 60 through 64. The benchmark is $\tau^k = 0.18$; Appendix D.1 explains the 0.36 alternative.

Fixing savings alone. A further experiment fixes the savings wedge at 0.009 while re-optimizing labor. The mean carbon wedge declines from 62.7% at 25 to 9.8% at 64 and -26.5% at 84, crossing the social cost of carbon between ages 41 and 42. Thus the declining price profile also occurs without imposing the benchmark labor schedule. The comparison uses the same 16×8 base grid and 40 shock nodes as its free-income control, with the additional marginal-utility grids adapted across regimes. It establishes a price-profile result; a corresponding savings-only no-carbon baseline has not been solved for a welfare comparison. If labor is fixed after savings, its sequential effect on retirement prices need not vanish: it changes the inherited promises and the savings multiplier even though the current labor-wedge constraint has no retirement content.

Alternative labor calibration. Table 5 compares the benchmark with the alternative labor anchors in Section 6.5. To summarize the age profiles, it applies common benchmark emissions weights across ages to each age’s unweighted mean wedge:

$$\frac{\sum_t \beta^t \bar{E}_t \bar{\tau}_t^q}{\sum_t \beta^t \bar{E}_t}.$$

Here $\bar{E}_t$ is mean emissions at age t in the benchmark allocation with both income wedges fixed, and $\bar{\tau}_t^q$ is the mean wedge in the allocation being summarized. Because households’ own emissions do not weight the within-age mean, this statistic is not carbon revenue per ton. Changes between successive restrictions depend on their order; the best linear rate is optimized separately.

Table 5: Fixed income wedges under two labor calibrations

	Benchmark schedules	GR16-implied anchors
<i>Effective carbon price (\$/tCO₂, common emission weights)</i>		
full optimum (nonlinear)	\$218	\$218
labor constraint	−49	+54
savings constraint	−62	−60
Both income wedges fixed	\$107	\$211
<i>The flat-rate restriction, within each constrained regime</i>		
best flat rate τ^{q*}	0.12 (\$47)	0.49 (\$187)
its welfare gain (%)	+0.021	+0.250
flat Pigouvian, $e = 0.494$ (%)	−0.115	+0.250
nonlinear wedge (%)	+1.401	+1.744

Note: the upper block uses common emissions weights across ages as defined above; the labor- and savings-constraint rows report changes between levels. The benchmark uses the normalized earnings schedule described in Appendix D.1. The lower block uses each column’s no-carbon baseline. Labor-only rows use their canonical grid; refining the alternative labor-only allocation raises its summary price by about \$10. Dollar rates convert at \$190 per ton = 0.494.

D.4 Welfare comparisons across calibrations

Table 6 gives the preference sensitivity behind Section 6.3. With income wedges free, its age-varying rule uses the full optimum’s mean wedge at each age and re-solves the income mechanism. The Stone-Geary column contains only linear prices because the unrestricted commodity solver requires homothetic preferences. The weak-separability welfare difference of 0.001 percentage points is below the numerical resolution described in Appendix D.7.

At the lower damage $e = 0.133$, the full optimum gains 0.0398% of consumption, the best linear rate 0.0376%, and the corrective rate 0.0355%, against that calibration’s no-carbon allocation. The best linear rate is 0.2027. These figures give the approximate welfare shares in Section 6.5.

At the higher capital tax $\tau^k = 0.36$, the best linear rate is 0.087 and gains 0.011%; the corrective rate loses 0.155%, while the nonlinear allocation gains 1.649%.

D.5 Age-dependent price families

Table 7 reports the searches behind Section 6.3.2. The candidate r_t takes the history-dependent allocation’s recorded mean wedge at each working age and a single β -weighted mean retirement rate. The first family scales deviations from the social cost of carbon: $e + a(r_t - e)$, with $a \in \{0, 0.75, 1, 1.25, 1.5\}$. The second scales the entire profile, mr_t , using eleven values of m from 0.25 to 2.5. The alternative shapes replace the working-life segment with a straight line

Table 6: Welfare value of carbon pricing with income wedges free, by preference calibration

	benchmark ($e = 0.494$)	Stone-Geary (necessity)	$\gamma_c = \gamma_q$ (placebo)	no externality ($e = 0$)
full optimum (nonlinear τ^q)	0.313	—	0.252	0.0041
age-dependent linear $\tau^q(t)$	0.312	—	—	0.0032
flat τ^{q*} (optimized)	0.311	0.152	—	0.0019
optimal flat rate τ^{q*}	0.56	0.56	—	0.038
flat Pigouvian $\tau^q = e$	0.309	0.150	0.253	—
emissions vs $\tau^q = 0$: full optimum	−15.0%	—	−13.5%	−1.8%
flat Pigouvian	−13.5%	−5.7%	−13.5%	—

Note: consumption-equivalent gains over each calibration’s no-carbon allocation, in percent. In the Stone-Geary column, the equivalent scales consumption above subsistence rather than total consumption (Appendix D.7). The optimal-rate row gives an ad-valorem rate; the last two rows give emissions changes. Dashes denote uncomputed Stone-Geary nonlinear and age-varying allocations, redundant intermediate rules under weak separability, or the corrective rule coinciding with the no-tax baseline when $e = 0$. Differences below 0.003 percentage points are unresolved.

between its endpoints or a constant rate at its emissions-weighted mean, then rescale to the same benchmark weighted mean rate before searching over m . All retirement segments are constant.

The floor experiment uses a revised candidate $\tilde{r}_t$: its working rates differ from r_t by at most 0.22 percentage points, and its retirement rate is -0.2200 instead of -0.2189 . The matched pair compares $\tilde{r}_t$ with $\max\{\tilde{r}_t, 0\}$. Figure 5 uses each schedule’s exported rates.

The welfare comparisons use the same 60,000 histories and recorded no-carbon baseline. The $a = 0$ check differs from the separately solved linear corrective allocation by 0.004 percentage points. Solving the same instrument through different warm-start sequences changes its reported welfare by about 0.016 points, so small differences between families are unresolved. In particular, the straight-line and level-scaled profiles are indistinguishable at this resolution. The coarser two-rate family loses appreciable value, but these restricted searches do not decompose the value of age variation into separate causal components.

The two-parameter family is $m[e + a(r_t - e)]$. The point with the highest computed welfare, $a = 1.25$ and $m \simeq 0.817$, gains 0.1538% at the lowest mean rate evaluated for that value of a . The maximum is not bracketed. Some attempted combinations implied a nonpositive retirement consumer price and were infeasible. Table 7 reports the largest gain among the evaluated schedules with complete welfare results; it does not claim an optimized age schedule. The matched floor comparison uses $m = 1$ in both runs, with abatement of 8.45% unfloored and 9.30% floored. Its approximately 0.02-point welfare difference is only modestly larger than

the observed variation across solution chains.

Table 7: Welfare gains across age-dependent carbon-price families

Age-price family	Parameter(s)	CE gain (%)
Scaling deviations from the SCC	$a = 1.25$	0.145
Level scaling	$m = 1.25$	0.124
Straight working profile	$m = 1.25$	0.126
Work and retirement blocks	$m = 0.75$	0.063
Scaling deviations and level	$a = 1.25, m = 0.817$	0.154
Matched profile, subsidies allowed	$m = 1$	0.129
Matched profile, zero floor	$m = 1$	0.109

Note: both income wedges are fixed at the benchmark schedules. Gains are in % of consumption over the same no-carbon allocation. The upper block reports the largest computed gain within each family and its associated parameter values; the lower block compares the candidate with and without a zero floor at the same scale. These searches cover restricted families, and the highest-welfare point in the two-parameter family lies at the search boundary. Differences of about 0.016 percentage points across solution chains are unresolved.

D.6 Implied transfers and labor-tax cuts

The transfer calculation in Section 6.4 uses the household's lifetime budget. Let $x_t = c_t + (1 + \tau_t^q)q_t$ denote spending at consumer prices. Under the normalized earnings schedule, retained earnings during working life are

$$H_t(y_t, \theta_{t-1}) = \theta_{t-1} \frac{\bar{\Lambda}_t}{1-p} \left(\frac{y_t}{\theta_{t-1}} \right)^{1-p}, \quad t < T_R,$$

with $H_t = 0$ in retirement. This integrates the net-of-tax marginal schedule with a zero intercept; $(1 - \bar{\tau}_t^y)y_t$ would give retained earnings only at $p = 0$. With zero initial and terminal assets, the implied lifetime transfer is

$$B(\theta^T) = \sum_{t=0}^T D_t [x_t - H_t], \quad D_0 = 1, \quad D_{t+1} = \frac{D_t}{R(1 - \bar{\tau}_t^s)}.$$

The discount factors use the household's after-tax return. The transfer increment ΔB and carbon payments $\sum_t D_t \tau_t^q q_t$ are computed on paired skill histories. The ratio of their means describes household present values, not a government revenue share discounted at R . The calculation identifies neither the timing of transfers nor a tax system that implements the allocation.

At the best linear rate, the mean transfer increment is 97.5% of mean discounted carbon payments, with a standard deviation of 2.4% of its mean. The observed variation is small, but its

economic component has not been separated from numerical error. A related zero-progressivity check produces residual transfer variation whose tilt changes sign with grid refinement. That diagnostic motivates caution about the small tilt; it supplies no error bound for the benchmark transfer increment. An exactly uniform increment has not been imposed and re-solved.

The labor-tax swap replaces $\bar{\Lambda}_t$ with $s\bar{\Lambda}_t$ and chooses s to keep $\mathbb{E}_0 B$ at its no-carbon level. This is the exercise’s mean-transfer neutrality condition. The savings wedge, and therefore the discount factors D_t , stay fixed, while individual transfers can change. Table 8 supplies the comparisons in Section 6.4.2. At rates 0.131 and 0.494, direct solves achieve mean-transfer neutrality to within 4×10^{-4} of discounted carbon payments. Other rows use the local linear relation in $\log s$ between the two solved allocations nearest neutrality; some require a short extrapolation. Setting $s = 1$ reproduces the dividend comparison within about 0.001 percentage points. The 0.8 dividend entry uses this check because no canonical dividend allocation is available there.

Table 8: The dividend and labor-rate-cut allocations along the carbon-rate grid

flat rate τ^q	$\approx \$/\text{tCO}_2$	dividend	swap s^*	rate cut (pp)	swap	swap – dividend
0.1	38	+0.021	1.0094	0.6	+0.144	+0.123
0.131	50	+0.020	1.0122	0.8	+0.182	+0.163
0.2	77	+0.014	1.0182	1.3	+0.254	+0.240
0.35	135	−0.035	1.0304	2.1	+0.367	+0.402
Pigouvian, $e = 0.494$	190	−0.115	1.0412	2.8	+0.432	+0.547
0.65	250	−0.226	1.0521	3.6	+0.465	+0.691
0.8	308	−0.357	1.0616	4.2	+0.453	+0.810

Note: welfare gains in % of consumption over the common no-carbon allocation; benchmark schedules and $\tau^k = 0.18$. The neutral scale s^* holds the mean implied transfer fixed. The rate cut is $(s^* - 1)(1 - \bar{\tau}^y)$ at the mean marginal rate, in percentage points. Rows at 0.131 and 0.494 have direct solves near neutrality; other rows use local linear approximation in $\log s$. Dollar rates convert at \$190 per ton = 0.494.

Under the alternative labor anchors, the corrective-rate swap gains about 0.286% of consumption against 0.250% for the dividend allocation. This is a near-neutral solved point: its mean transfer falls short of the no-carbon level by 1.3% of discounted carbon payments. Its decile mean gains decline from 0.47% at the bottom to 0.16% at the top, reversing the benchmark gradient. These comparisons retain the allocation and implementation qualifications of Section 5.1.2.

D.7 Numerical methods and validation

The solvers use backward induction on multipliers dual to promise keeping and threat keeping, following Farhi and Werning (2013). With income wedges free, an adaptive Dormand-Prince

5(4) scheme integrates the within-period boundary-value problem in type, at relative tolerance 10^{-3} and absolute tolerance 10^{-6} ; Brent’s method closes it by shooting on the bottom promised value. The constrained solvers add the marginal-utility state $\hat{M}$ of Lemma 5.1 and use trapezoid collocation with a sparse analytic Newton Jacobian.

The finer welfare comparison uses 100×45 dual grids and 20 type nodes. The constrained calculations use 16×8 dual grids, 40 type nodes, and 11 to 27 marginal-utility nodes. The matched free-income rate sweep in Figure 4 also uses the 16×8 grid and 40 type nodes. It gives free-income welfare gains of 0.315% at the best linear rate and 0.311% at the corrective rate, slightly different from the finer-grid values in Table 6. Comparisons retain their own no-carbon baseline.

Aggregate welfare uses mean simulated lifetime utility, including retirement. If $\Delta v(\theta^T)$ is a history’s utility change relative to the comparison’s no-carbon allocation, the reported gain is

$$g = \exp\left(\frac{\mathbb{E}_0[\Delta v]}{\mathcal{A}}\right) - 1, \quad \mathcal{A} = \frac{1 - \beta^{60}}{1 - \beta}. \quad (45)$$

Tables and figures report $100g$. In the homothetic benchmark, g is the common proportional increase in both baseline consumption goods that gives the same mean utility gain with labor fixed. In the Stone-Geary variant, the formula instead scales c and $q - \bar{q}$, or expenditure net of subsistence at fixed prices. It is not a percentage increase in total consumption. The incidence tables and decile comparisons average individual equivalents $\exp(\Delta v/\mathcal{A}) - 1$ within each group; because the conversion is nonlinear, their population mean need not equal (45).

Agent-specific random-number streams keep skill histories common across compared regimes. The finer welfare comparison uses 300,000 histories and the matched sweeps use 60,000. In the former, the spread in present-value resource residuals is about 0.003 percentage points of consumption, versus a seed-to-seed standard deviation of gains of 0.0003 points. This is a resolution diagnostic rather than a bound on total error; smaller welfare differences are unresolved. Appendix D.5 reports the larger variation observed across solution chains in the age-price searches.

The consistency checks cover analytic derivatives against finite differences (errors between 10^{-5} and 10^{-9}), the one-good replication, and nested policy regimes. With the two-good extension switched off, policies agree with the replication to 4×10^{-3} and the initial resource multiplier to 2.6×10^{-4} ; tighter integration tolerances reduce these gaps by over two orders of magnitude. With $\varkappa = 0$, the carbon wedge equals marginal damage to 2.8×10^{-16} at the evaluated grid points. Fixing income wedges at their desired counterparts approaches the free-income solution as the type grid is refined: the initial-multiplier gap falls from 0.012 to 0.0038 when the type grid doubles from 20 to 40 nodes.

The full-optimum carbon premium’s approximation $\varkappa l^\alpha \tau^y / (1 - \tau^y)$ differs from the solved

premium normalized by $1 + \tau^q$ by at most 2.1% across ages. Optimal labor wedges remain close to the one-good values: 0.021 to 0.363, against 0.021 to 0.367. The savings wedge falls from 0.56% at entry to 0.02% at age 63.

One diagnostic remains unresolved. Off the storage grid, reconstructing the income-wedge constraint multipliers leaves a mean absolute discrepancy of about 0.18 between the node-condition reconstruction and the solved carbon-price deviation; collecting the same inputs through the manuscript's formula adds a discrepancy of 0.024. Tight residuals at storage nodes do not resolve this mismatch. The reported carbon-price profiles and comparisons that impose the constraints sequentially use solved allocations, not the reconstructed multiplier decomposition.